

From Data to Models: Analyzing and Integrating Biological Data into Mechanistic Models

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Course of *Doctoral school*
PhD in Complex Systems for Quantitative Biomedicine

When?

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
14	15	16	17	18	19	20
					• 09:00 - Seminar	
21	22	23	24	25	26	27
• 09:00 - conference, terzo piano					• 09:00 - 13:00 Sala conferenze, terzo piano	
28	29	30	1	2	3	4
• 09:00 - 13:00 Sala conferenze, terzo piano						

DAY 3: Outline

From Case Study to Data Analysis

- From Data to Models: the importance of model parameterization using real (raw) experimental data.
- Real-World Case Study: understanding the problem context.
- Real-World Case Study: model definition.
- Real-World Case Study: model calibration.



<https://github.com/qBioTurin/From-Data-to-Models.git>

Background: cell polarity

Cell polarity: organized asymmetry

All eukaryotic cells can acquire transient cell polarity.

Some eukaryotic cells are intrinsically polar:

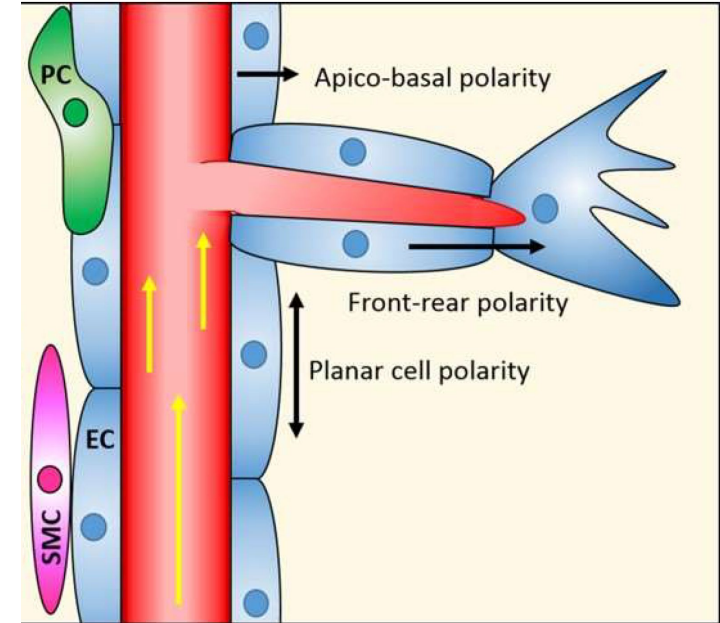
- simple epithelium of the gut lining and kidneys
- neurons
- endothelial cells



Three types of polarity axes:

1. transient front-rear polarity (during sprouting)
2. planar cell polarity (due to blood flow)
3. apico-basal polarity

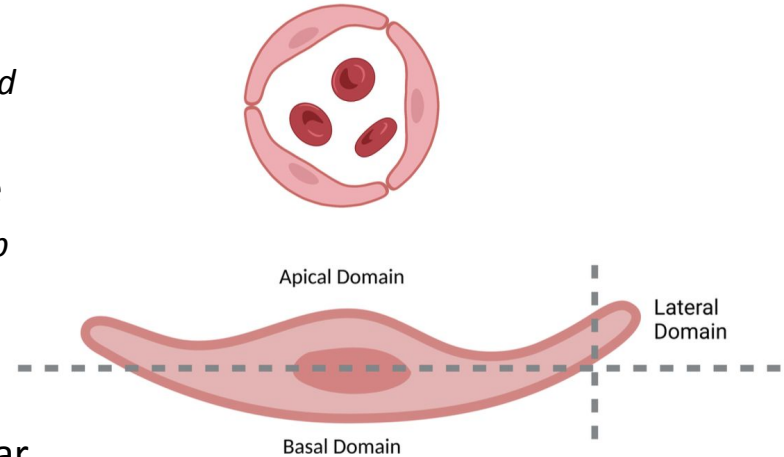
Cell polarity is a morphological and functional identity.



Background: apico-basal polarity

Apico-basal polarity defines 3 morphological and functional domains:

1. **apical domain:** in direct contact with blood flow
morphology: glycocalyx and soluble molecules
function: regulation of vasoconstriction/dilation, coagulation and inflammation
2. **lateral domain:** where inter-endothelial junctions reside
morphology: inter-endothelial junctions (tight, adherens and gap junctions)
function: connect adjacent cells + molecular barrier that ensures maintenance of polarity
3. **basal domain:** which anchors the cells to the extracellular matrix (ECM)
morphology: several adhesion molecules, including integrins
function: anchoring the cell to the basal membrane (BM) and ECM, which is in continuous turnover



Background: integrins

Integrins are *heterodimers* of non-covalently associated α and β subunits.

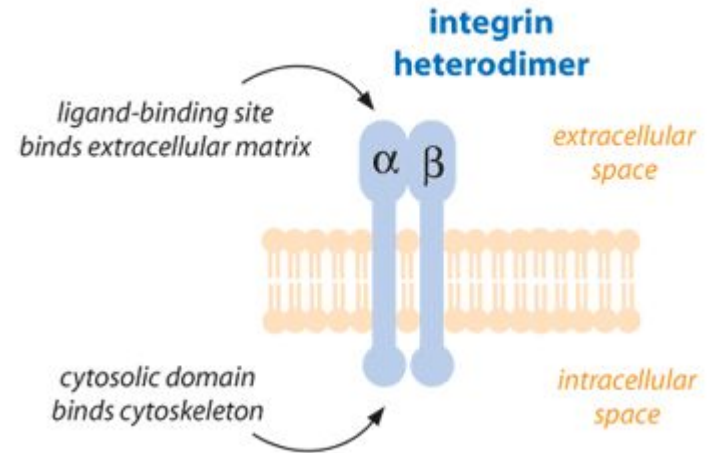
24 four distinct receptors can be assembled through different combinations of:

18 α subunits and

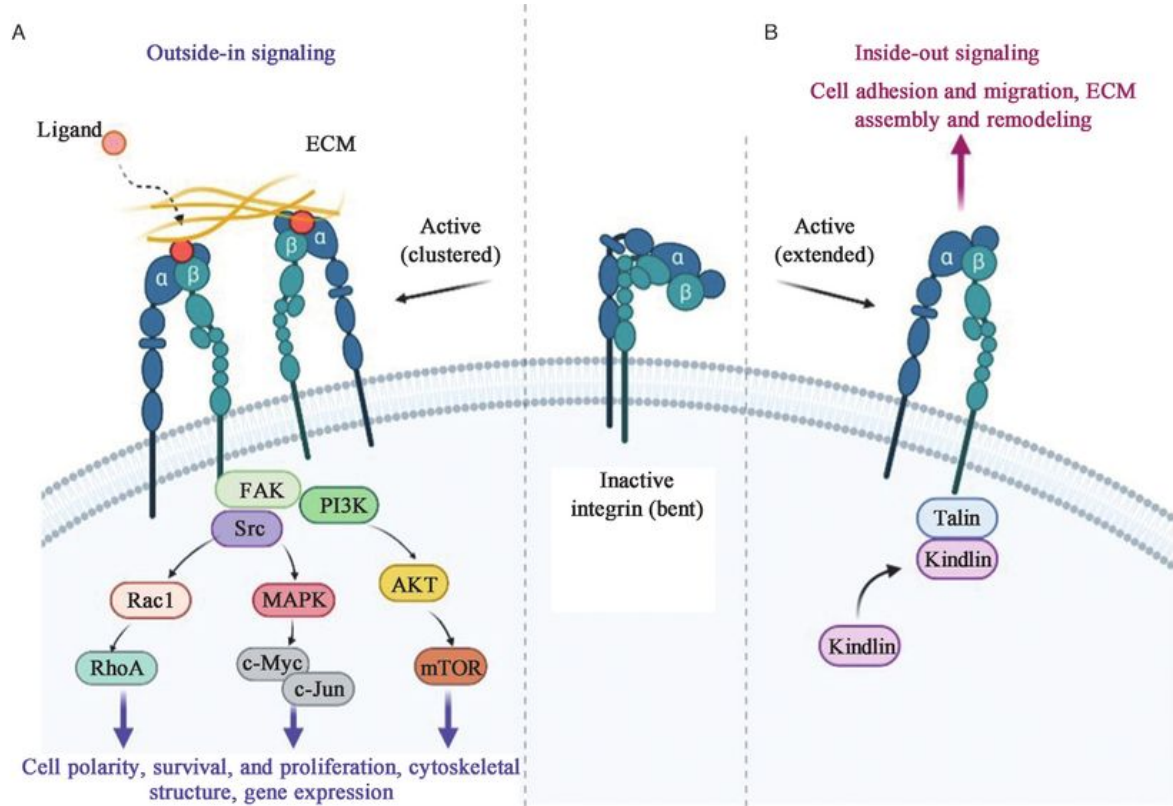
8 β subunits,

each resulting receptor with specific binding properties and nonredundant functions

Their structure enables them to mediate bidirectional communication (***dynamic reciprocity***)



Background: outside-in / inside-out signaling

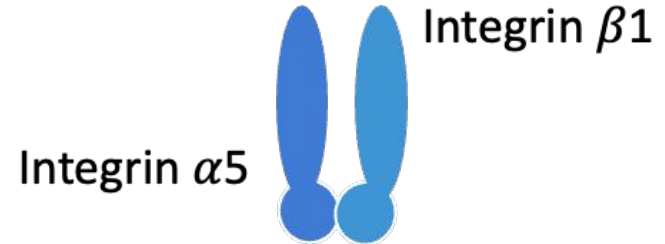


Background: integrin $\alpha 5 \beta 1$ and fibronectin1

Fibronectin is a multi-domain glycoprotein that exists in two distinct forms:

1. plasma FN, which is a soluble, is secreted by hepatocytes and circulates in the bloodstream to mediate hemostasis upon tissue injury
2. cellular FN: insoluble fibrillar, secreted by ECs and requires a cell-mediated process to be polymerized within the matrix

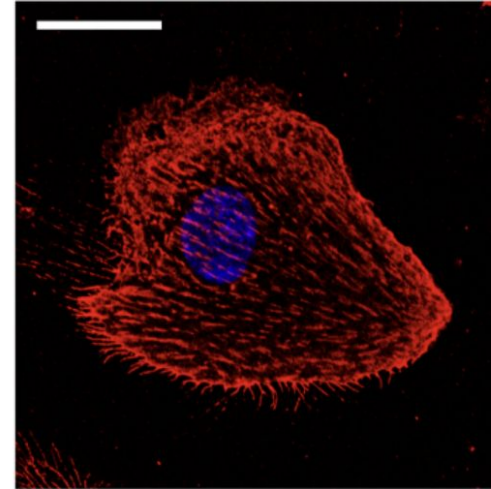
Both plasma and cellular FN are encoded by a single gene, and they are secreted as dimers of two similar yet not identical polypeptide chains



Background: active integrin $\alpha 5 \beta 1$ localizes on basolateral membrane

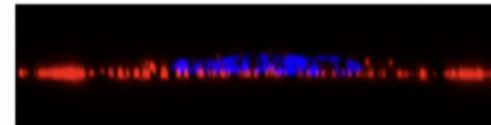
Confocal xy and xz sectioning microscopy analysis of anti-**active $\alpha 5 \beta 1$ integrin** localization on the cell surface of living human umbelical arthery endothelial cells (HUAECs). Active $\alpha 5 \beta 1$ integrin is selectively distributed on basal EC membrane. **Nuclear staining** with DAPI

XY

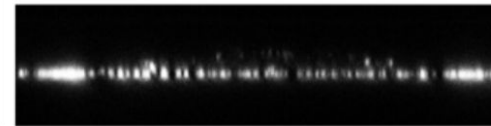


Active $\alpha 5 \beta 1$
DAPI

XZ



Active $\alpha 5 \beta 1$
DAPI

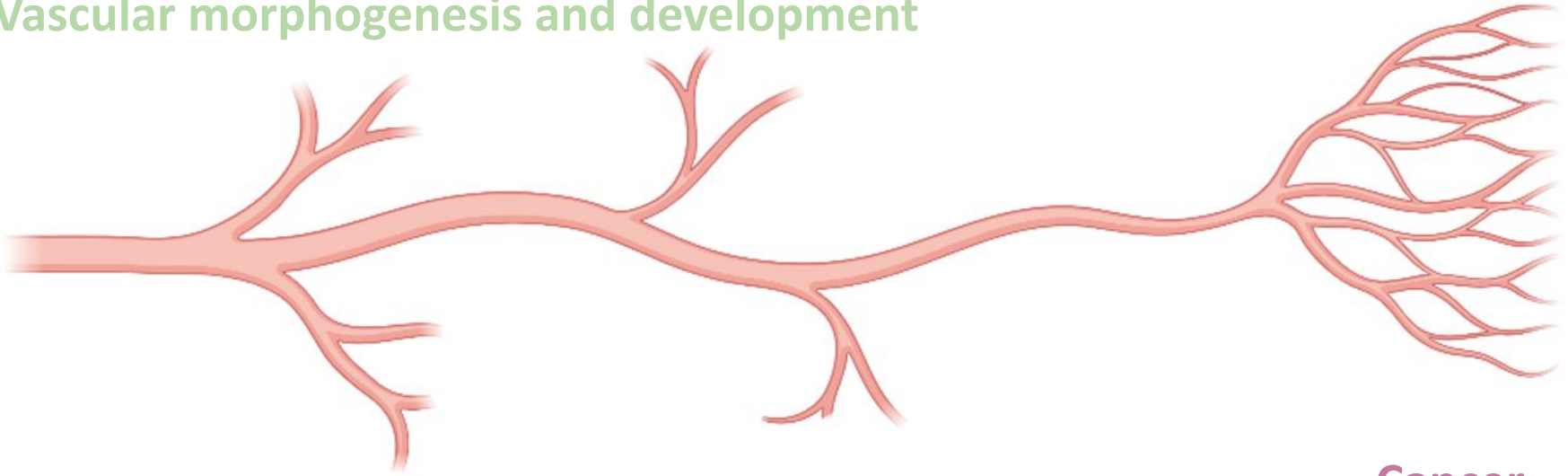


Active $\alpha 5 \beta 1$

Real world case study: endothelial cell polarity

Endothelial cell polarity is **crucial**
for physiological and pathological processes

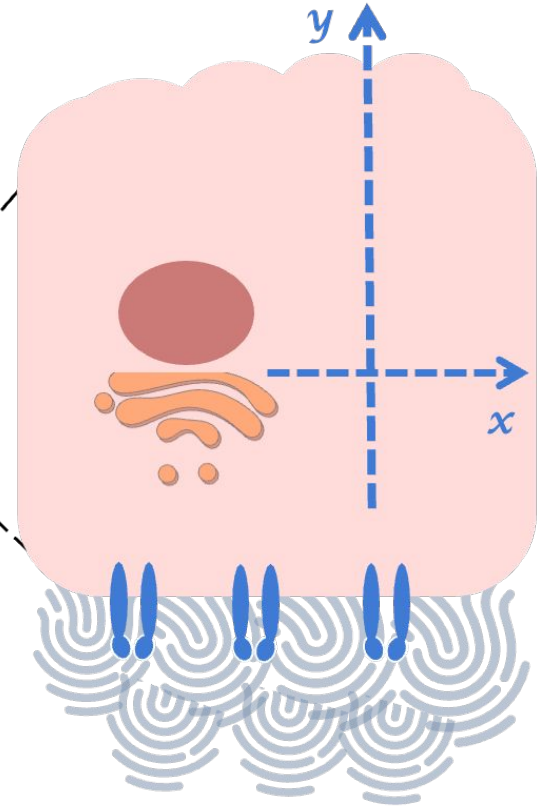
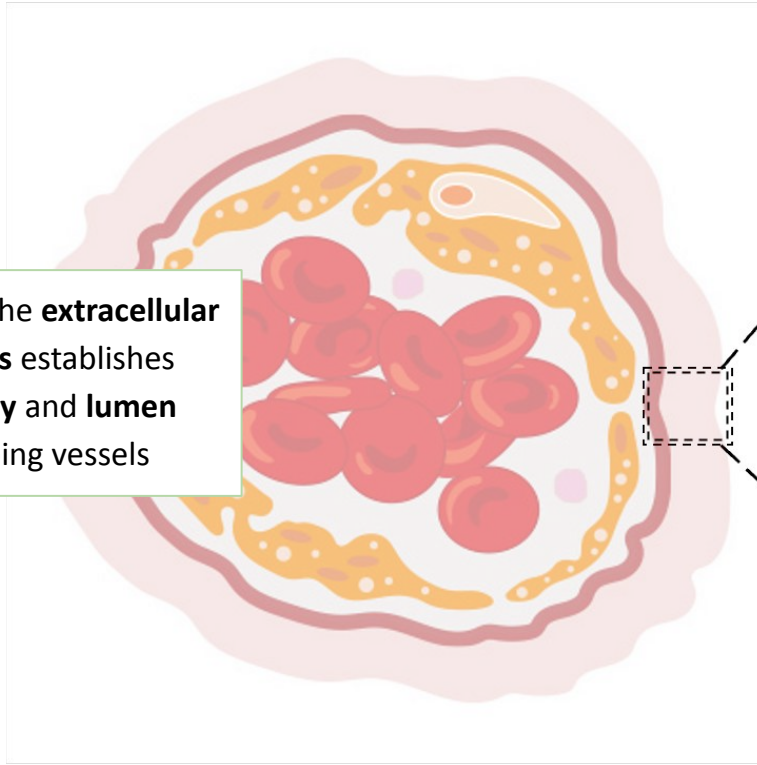
Vascular morphogenesis and development



Cancer

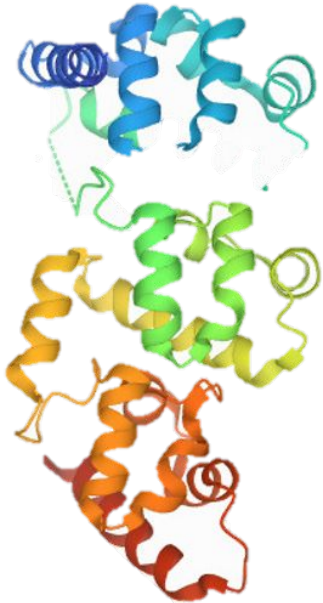
Maintenance of endothelial cell polarity

The interaction between the **extracellular matrix** and **$\beta 1$ integrins** establishes **endothelial cell polarity** and **lumen formation** in developing vessels

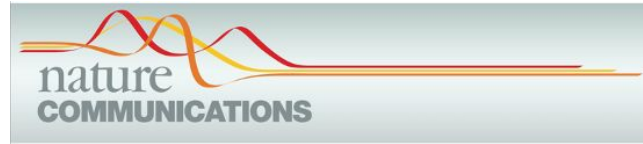


Real world case study: endothelial cell polarity

PTPRF-interacting protein $\alpha 1$ (PPFIA1) is a key mediator of EC polarity



Crystal structure of
PPFIA1 SAM domain



ARTICLE

Received 23 Aug 2015 | Accepted 13 Oct 2016 | Published 23 Nov 2016

DOI: [10.1038/ncomms13546](https://doi.org/10.1038/ncomms13546)

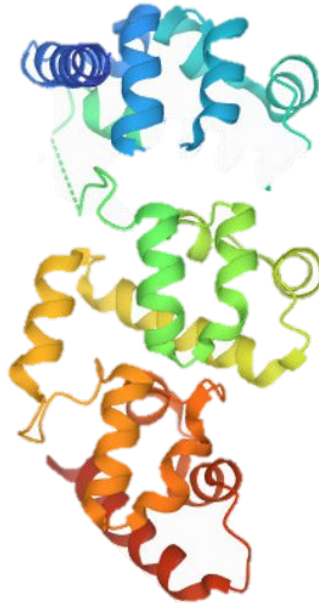
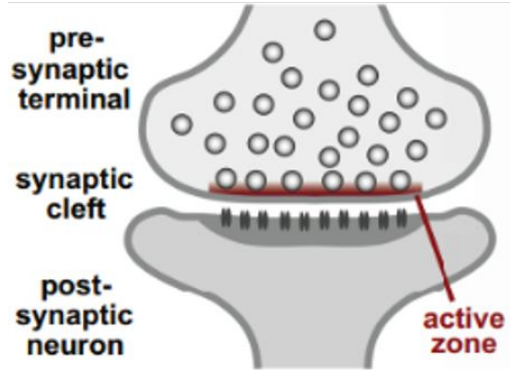
OPEN

PPFIA1 drives active $\alpha 5 \beta 1$ integrin recycling and controls fibronectin fibrillogenesis and vascular morphogenesis

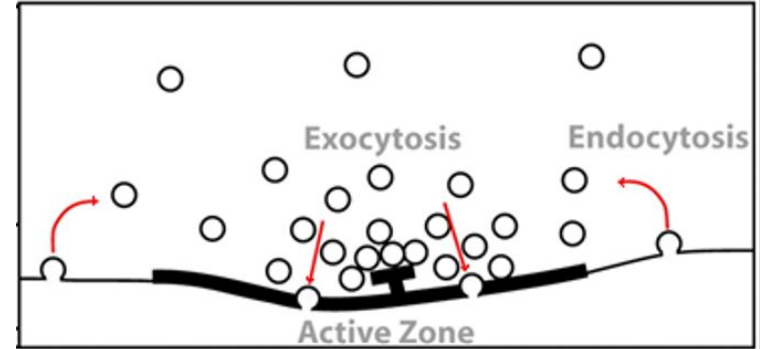
Giulia Mana^{1,2,*}, Fabiana Clapero^{1,2,*}, Emiliano Panieri³, Valentina Panero³, Ralph T. Böttcher⁴, Hui-Yuan Tseng⁴, Federico Saltarin^{1,2}, Elena Astanina^{1,5}, Katarzyna I. Wolanska⁶, Mark R. Morgan⁶, Martin J. Humphries⁷, Massimo M. Santoro^{3,8}, Guido Serini^{1,2,**} & Donatella Valdembri^{1,2,**}

Real world case study: endothelial cell polarity

PPFIA1 docks neurotransmitter-containing vesicles in synapses

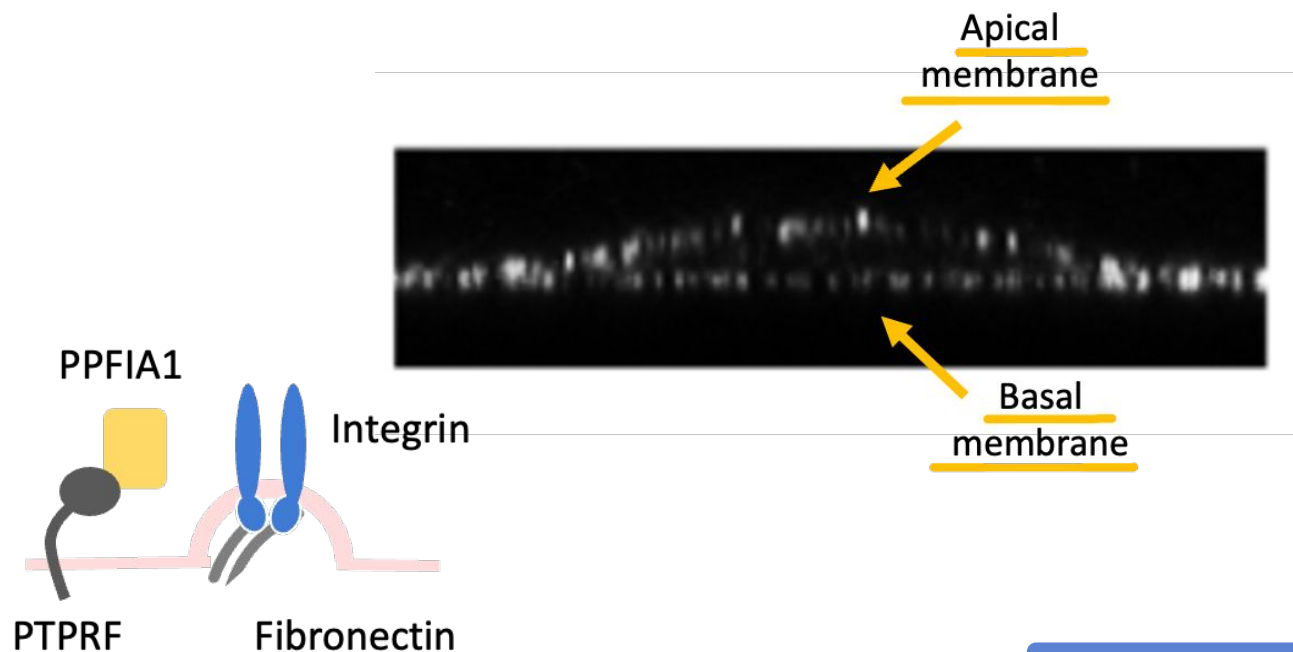


Crystal structure of
PPFIA1 SAM domain



Real world case study: endothelial cell polarity

PPFIA1 docks vesicles containing cellular adhesion receptors on the basolateral membrane of ECs



Cellular model:
Human Umbelical Aortic
Endothelial Cells (HUAECs)

Polarized cell



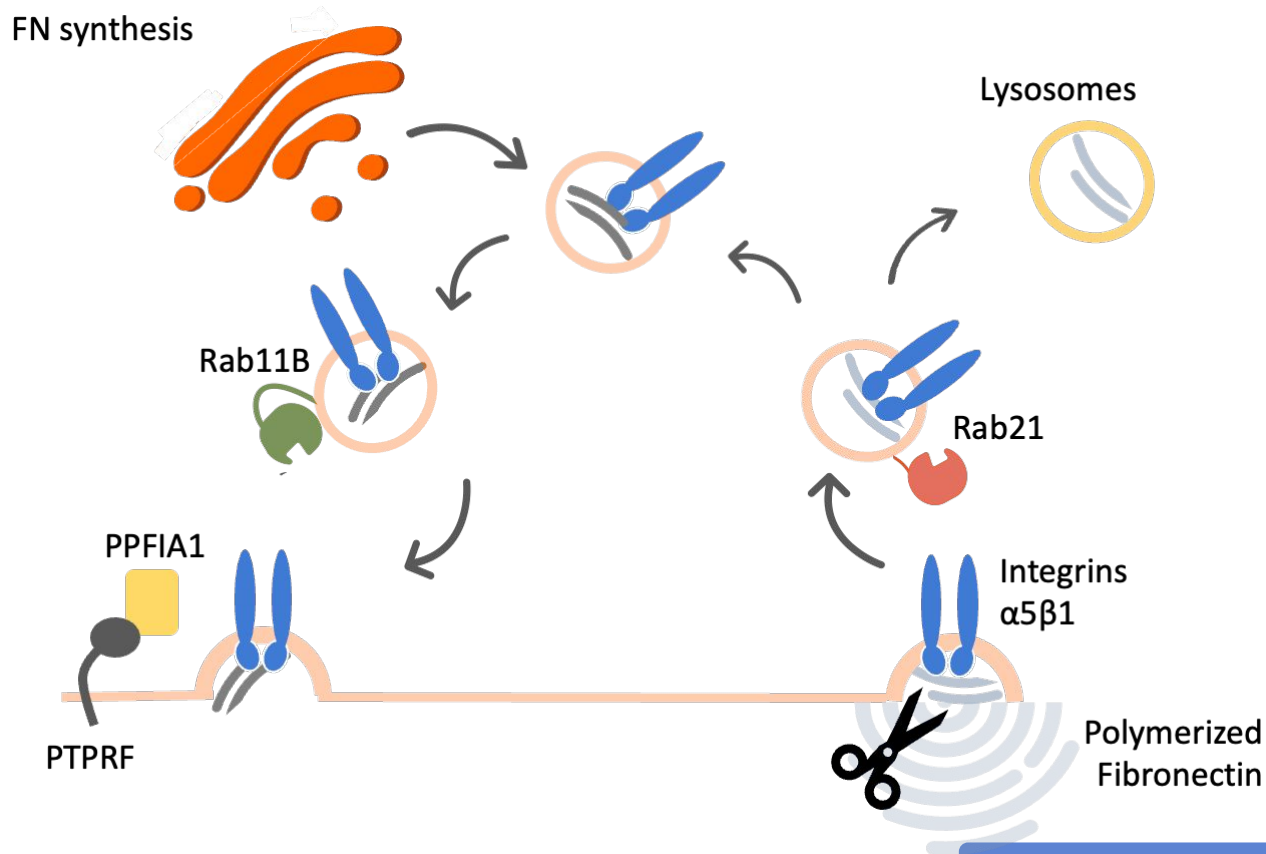
Integrin distributed on the membrane in CONTROL cells

NON - polarized cell

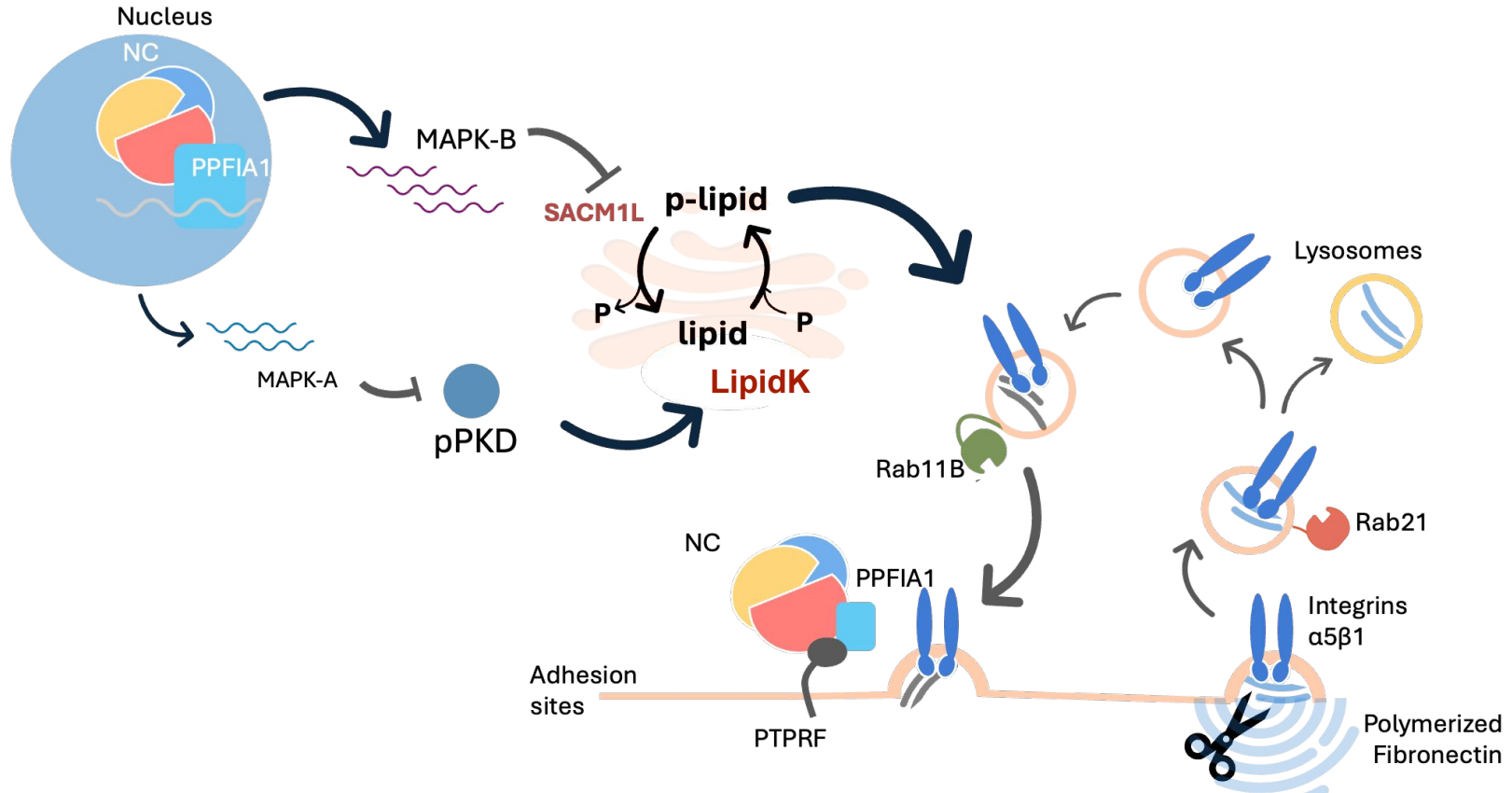


Integrin distributed on the membrane
in PPFA1 KNOCK-OUT cells

PPFIA1-dependent pathway: integrin recycling



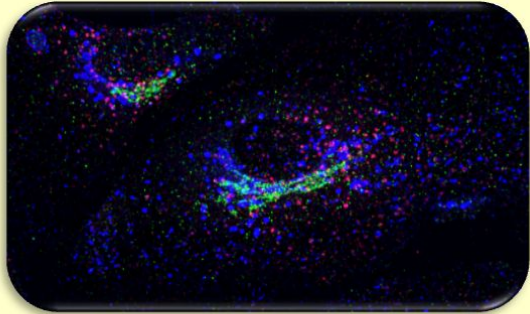
PPFIA1-dependent pathway: Golgi-secretion regulation



Research outline

Cellular Biology approach

Immunofluorescence
Live imaging
Western Blot
Real-time PCR

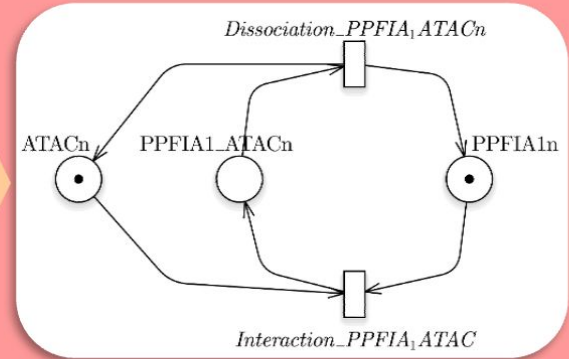


Understanding at fine **molecular detail**
the **components** involved in the pathway

Data-driven model

Computational Biology approach

Petri Net formalism



Viewing the system **as a whole**
and investigating its characteristics

Translating biological knowledge into a computational model

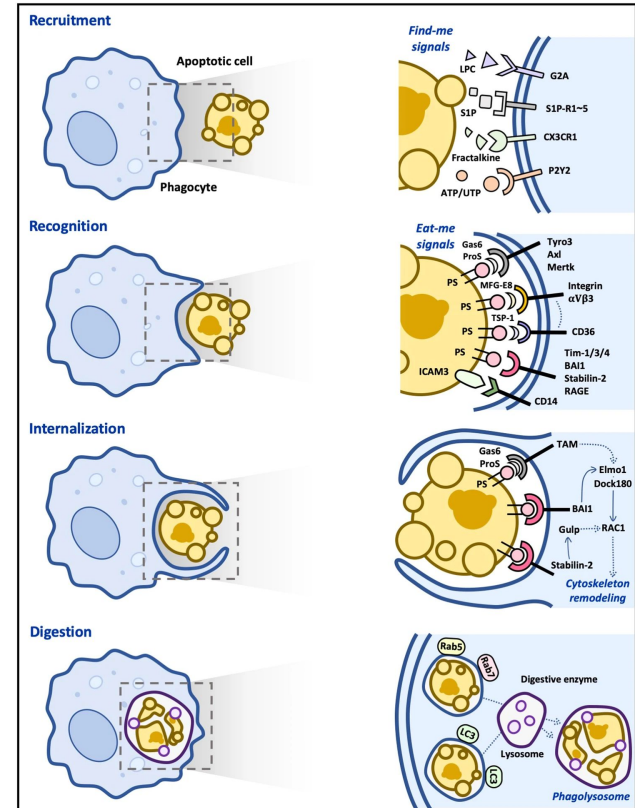
Model construction:

- translating biological graphical models into formal models
- multi-scale complexity: different levels of complexity and how they might interact across scales and levels of organization

Data integration: models are dependent on quantitative data (e.g. lab experiments, omics, imaging)

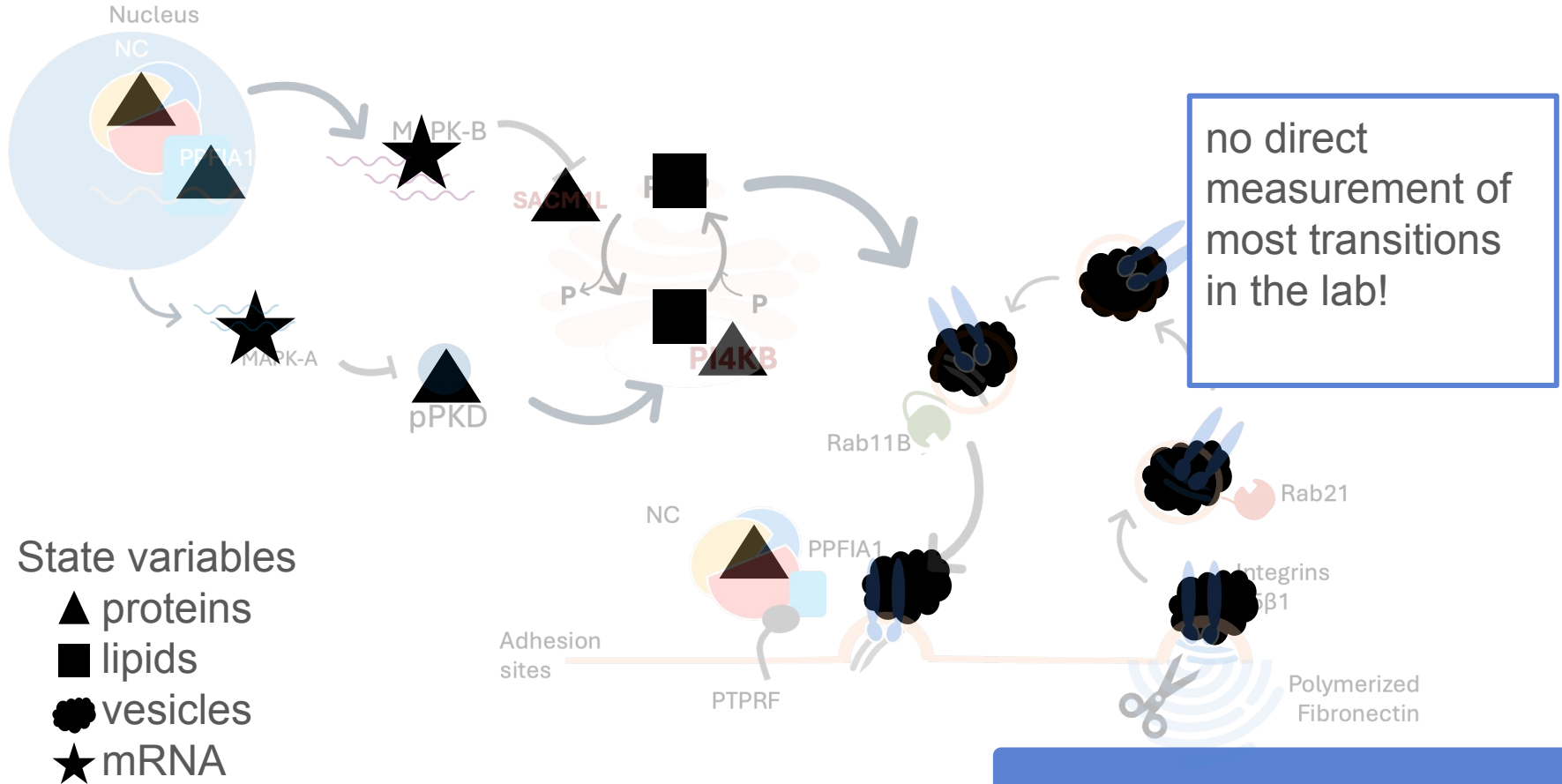
data is:

- required for parameter estimation, validation and prediction
- sparse and noisy, with limited time points
- can come from heterogeneous sources (in-house and from the literature) → can you compare it?
- can come from different types of experiments of different types of molecules → can you merge different types measurement units?
- is often a relative abundance data, but models require absolute values
- some data is unavailable and must be estimated (kinetic rates, binding constants)

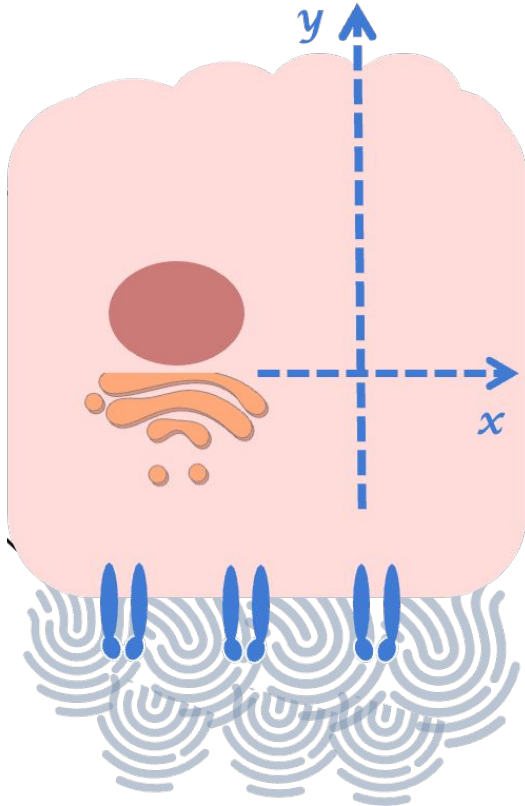


Byeongjin Moon et al., *Experimental & Molecular Medicine*, 2023

PPFIA1-dependent pathway overview



What are we modeling and why?



We are modeling a single endothelial cell.

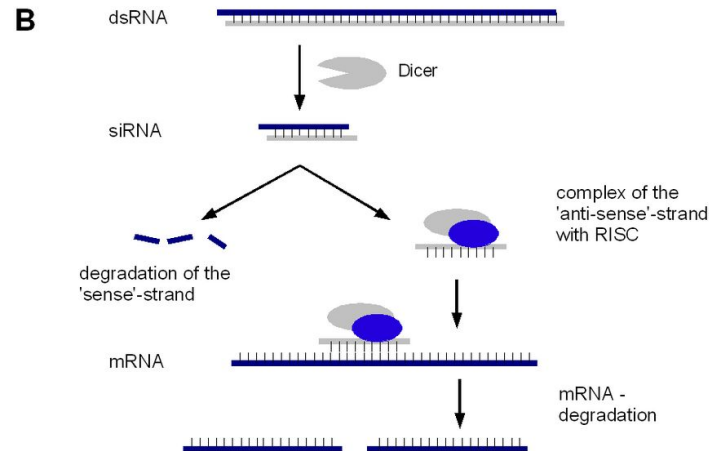
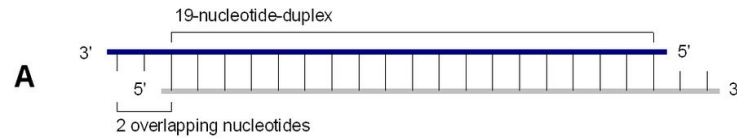
What happens to cellular polarity as PPFIA1 is progressively removed from the cell?

We want to:

- explore the dynamics of the system
- identify gaps in our knowledge
- test hypotheses *in silico* first and validate *in vitro*

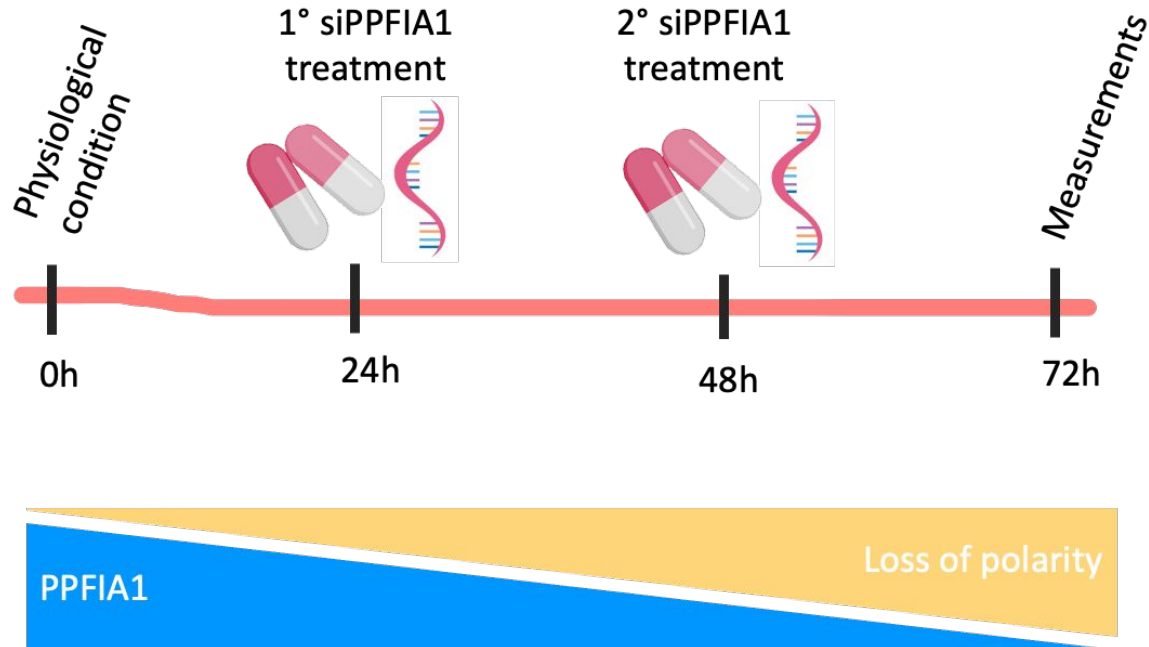
What are we modeling and why?

We are modeling a single endothelial cell as
PPFIA1 is progressively removed by siRNA.



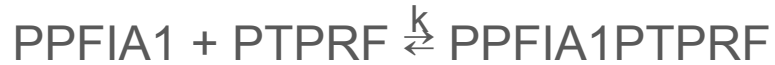
What are we modeling and why?

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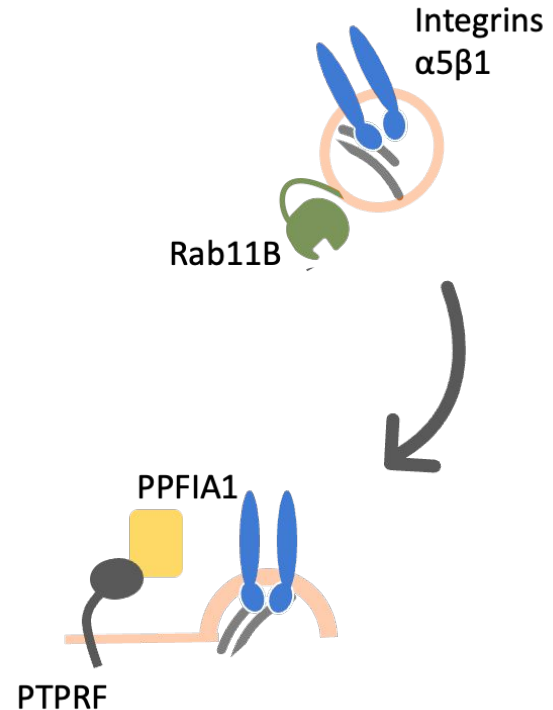


First step: drawing the Petri Net from biological knowledge

1. Gather biological knowledge
2. Write events as biochemical reactions



3. Do I have all necessary information to fully parametrize the reaction (T0 and Tend)?
 - stoichiometry
 - velocity of reaction
4. Can I find them in the literature?
5. Should I redefine level of abstraction?

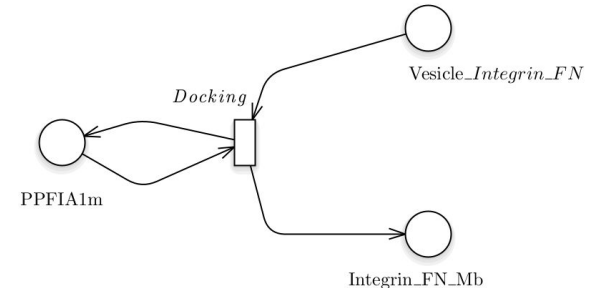
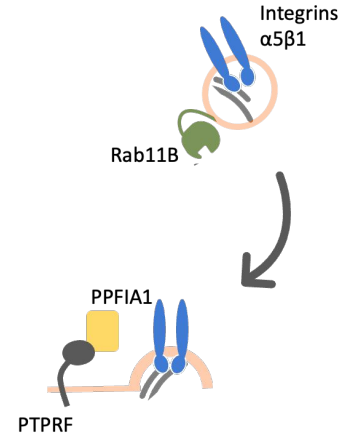


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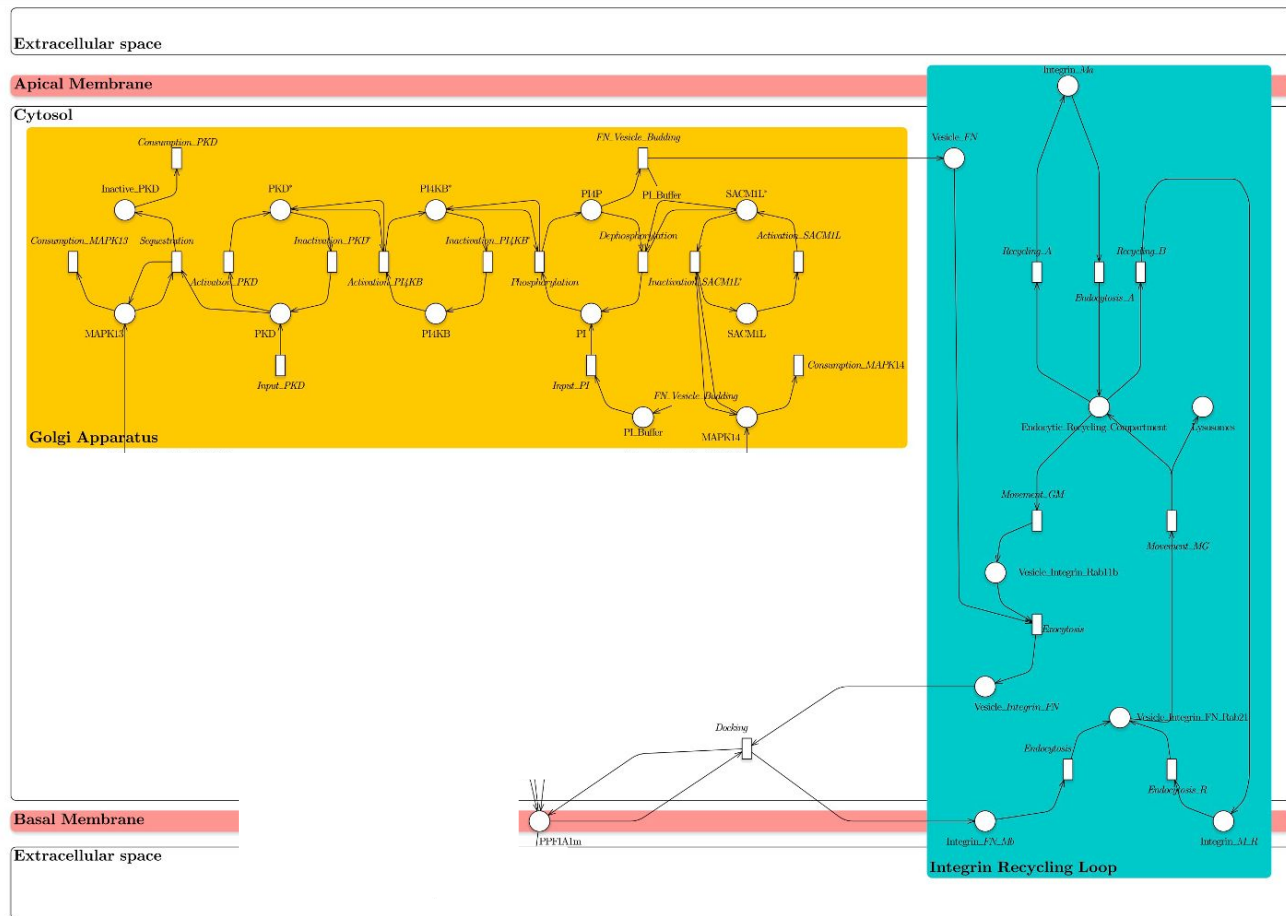
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First step: drawing the Petri Net from biological knowledge



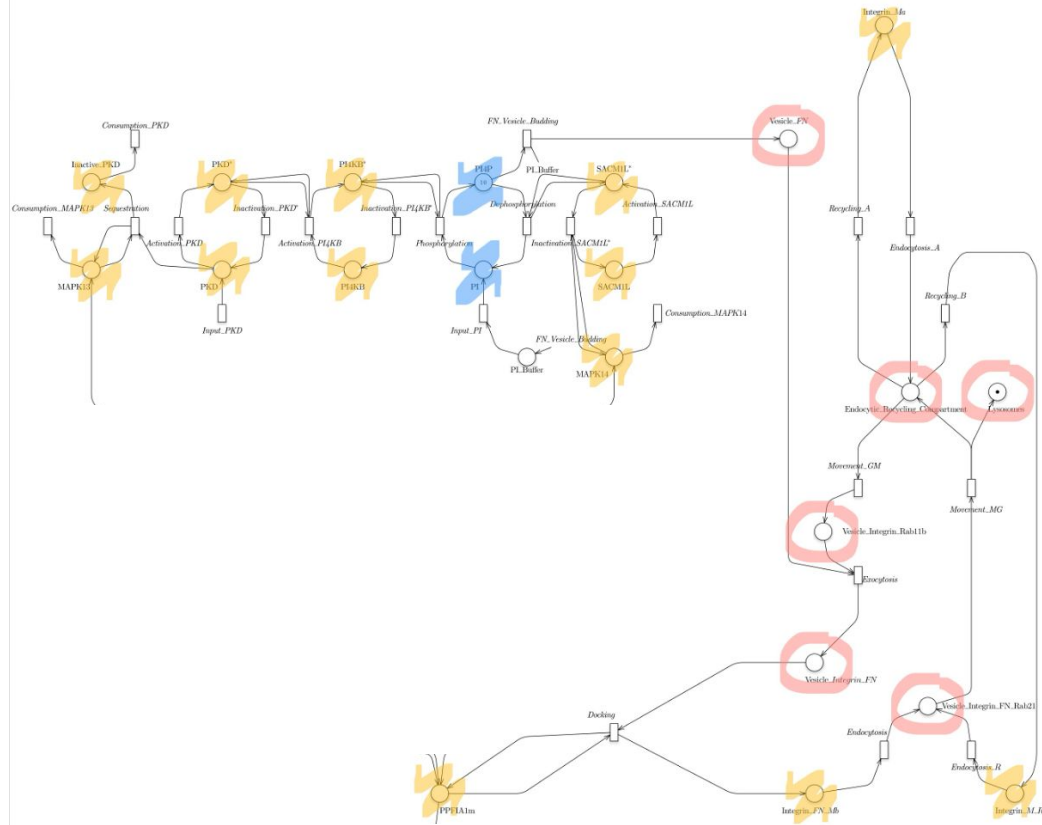
Second step: how can I integrate experiments?

State variables:

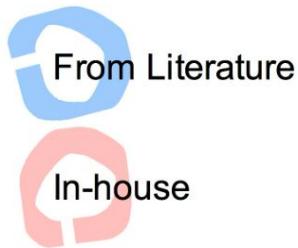
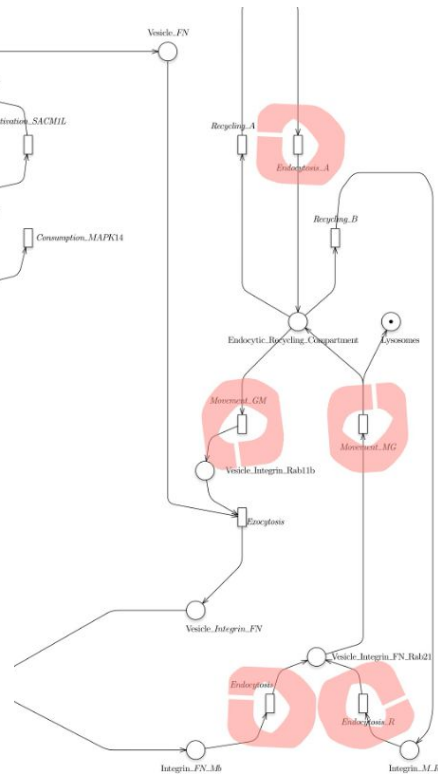
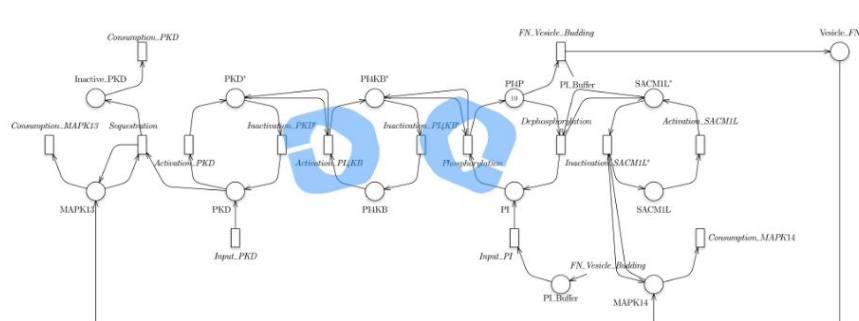
Proteins

Lipids

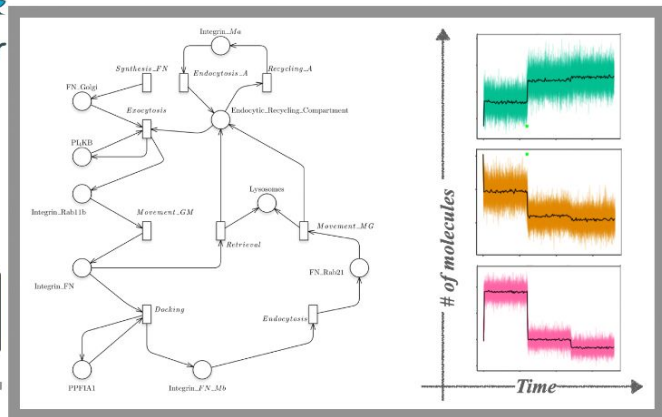
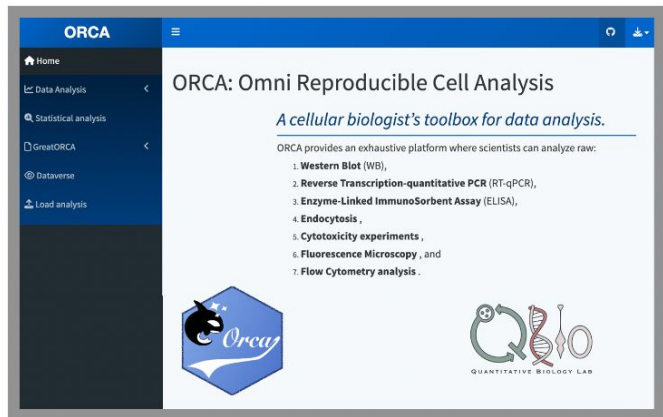
Vesicles



Second step: how can I integrate experiments?



GreatORCA



The
Dataverse[®]
Project

GReatOrca

GReatMod

OmniReproducibleCellAnalysis





GreatORCA combines:

- data analysis of most common lab experiments
- data integration into computational models

ORCA purposes



Open Science

Sharing of research methods, outputs and data freely to enhance collaboration and knowledge dissemination



Reproducibility

Scientific results can be consistently replicated by independent researchers using the same methodology and data



Transparency

Clear and open communication of all aspects of the research process, including methodologies, data, and analyses



Compiles Report

Download Report



Standardization

Establishing uniform protocols and guidelines to ensure consistency, accuracy, and comparability of results across studies



User-friendly

Approachable by scientists without advanced R language knowledge



❖ Analysis of wet lab experiments

- ➡ ● Western Blot (**WB**),
- ➡ ● Reverse Transcription-quantitative PCR (**RT-qPCR**),
- Enzyme-Linked ImmunoSorbent Assay (**ELISA**),
- Endocytosis
- Cytotoxicity
- ➡ ● Fluorescence microscopy
- Flow Cytometry analysis experiments (**FACS**)
- ... and growing

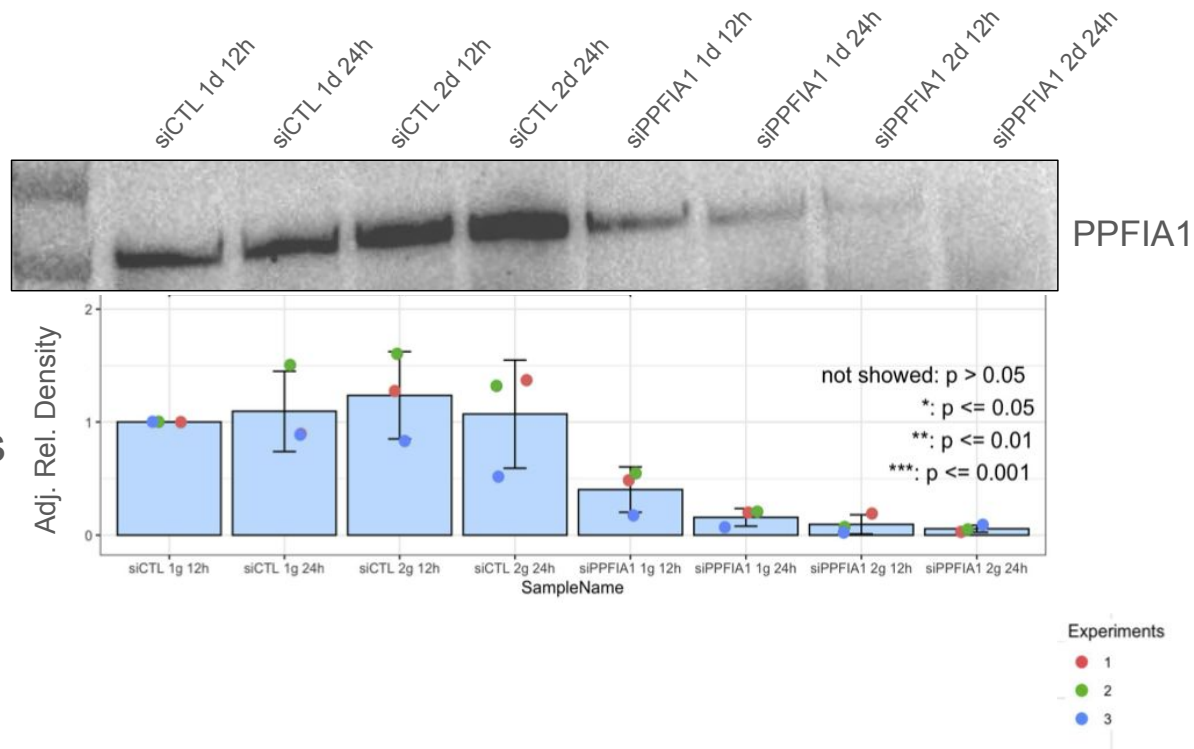
➡ ❖ Support for statistical analysis

➡ ❖ Integration of wet lab experiment results in a computational model

Data integration: relative abundance vs absolute abundance

Most wet lab experiments produce results as relative abundances.

Tokens in Petri Nets represent entities and must be defined as absolute quantities



Data integration: relative abundance vs absolute abundance

Most wet lab experiments produce results as relative abundances.

Tokens in Petri Nets represent entities and must be defined as absolute quantities.

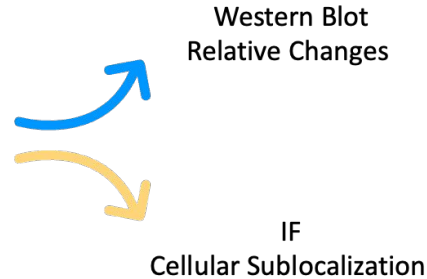
Strategy: identify experiments with absolute quantifications and use it to convert relative abundances to absolute abundances



Model parametrization – Proteins

Mass spectrometry (iBAQ)
Madugudungu et al., *Proteomics*, 2019

$$\text{iBAQ} = \frac{\sum \text{peak intensity}}{\# \text{ theoretical peptides}}$$



Data integration

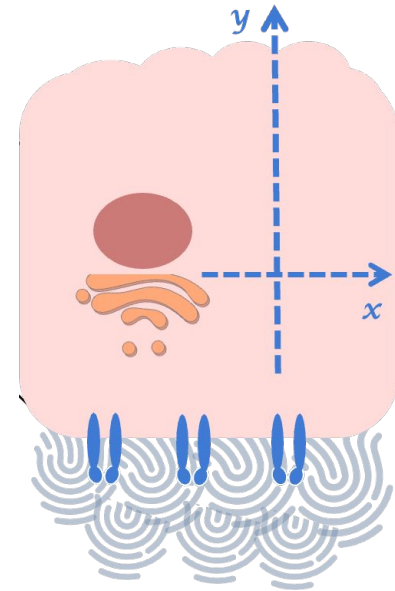
Objective: We are modeling a **single endothelial cell** as PPFIA1 is progressively removed by siRNA.

Mass Spec experiment
Madugunu et al., *Proteomics* 2019



10^5 HUVECs

iBAQ/ 10^5



Example: amount of PPFIA1

Statistica complessiva

	Rosso	Blu	Nero	Media	SD
siCTL 1gg 24h	1	1	1	1.00	0
siCTL 2gg 24h	2.2250136	0.745316	0.620383	1.20	0.892557
siPPFIA1 1gg 24h	0.1821956	0.154055	0.079566	0.14	0.05303
siPPFIA1 2gg 24h	0	0	0	0.00	0

	siCTL 1gg 24h	siCTL 2gg 24h	siPPFIA1 1g	siPPFIA1 2gg 24h
Media	1.00	1.20	0.14	0.00
SD	0.00	0.89	0.05	0.00

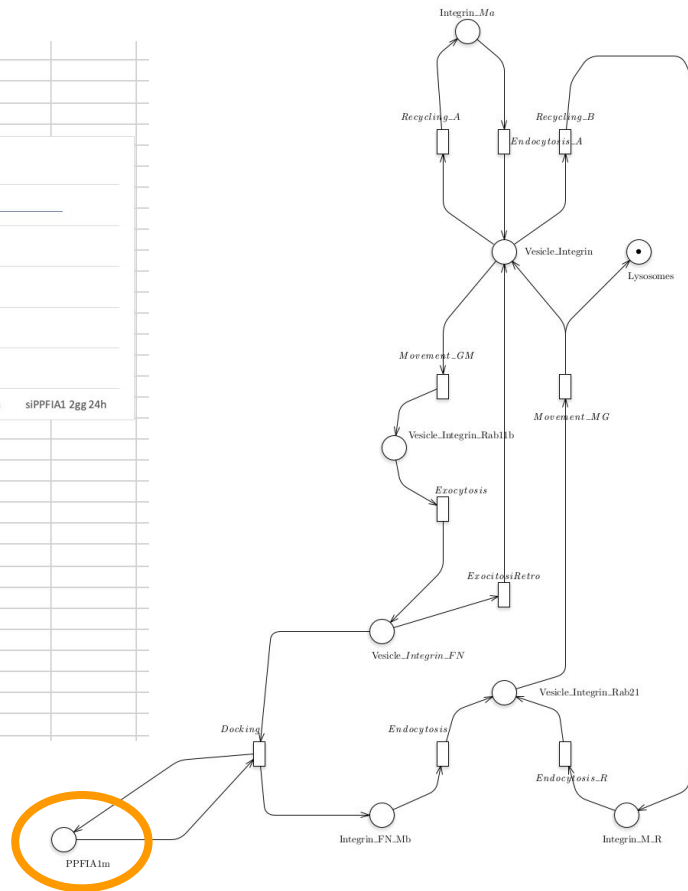
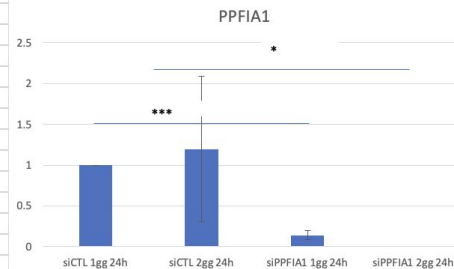
TESTT	4.748E-06 ***
	0.0404505 *

Da Madugundu 2019 normalizzato (vedi file Mass Spec), ottengo numero assoluto di quantità di proteina:
3.216359226

Quindi:	siCTL 1gg 24h	siCTL 2gg 24h	siPPFIA1 1g	siPPFIA1 2gg 24h
	3.2163592	3.849674	0.445805	0

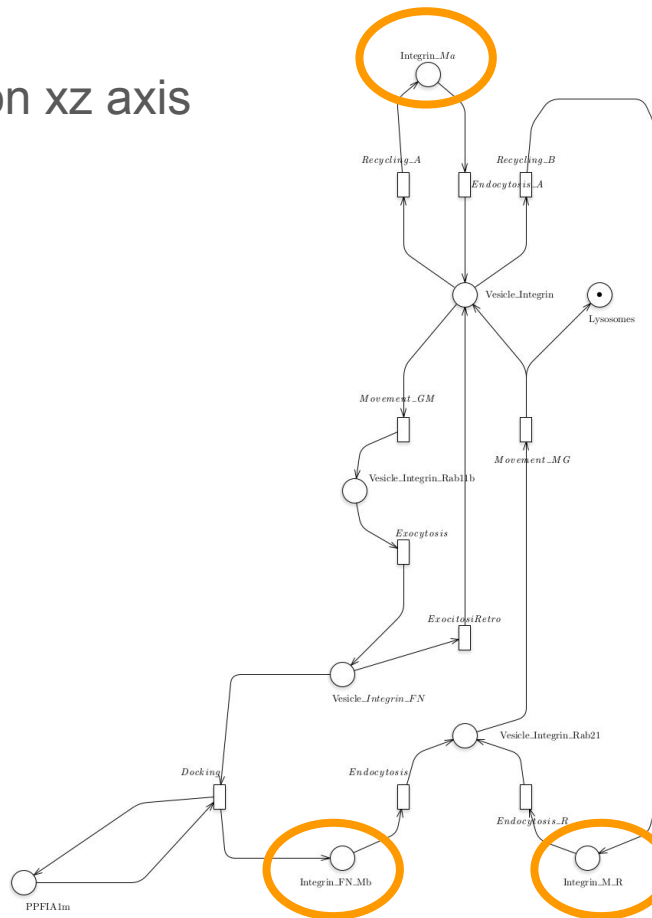
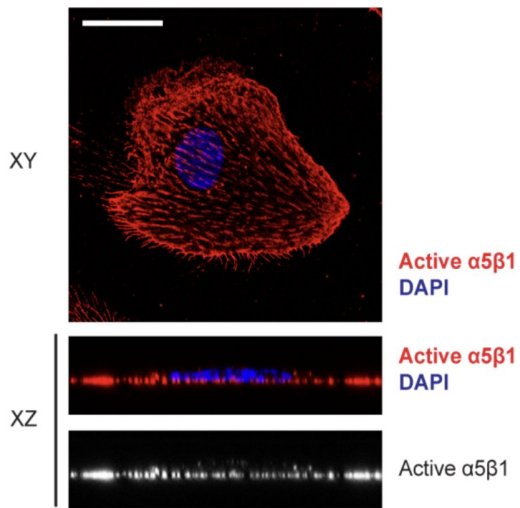
Quindi nel modello

	corrisponde a tempo 0h	0h	24h	48h	72h
siCTL 1gg 24h					
no dato per	tempo 24h		3.216359226	0.445805	0
siPPFIA1 1gg 24h	corrisponde a tempo 48h				
siPPFIA1 2gg 24h	corrisponde a tempo 72h				



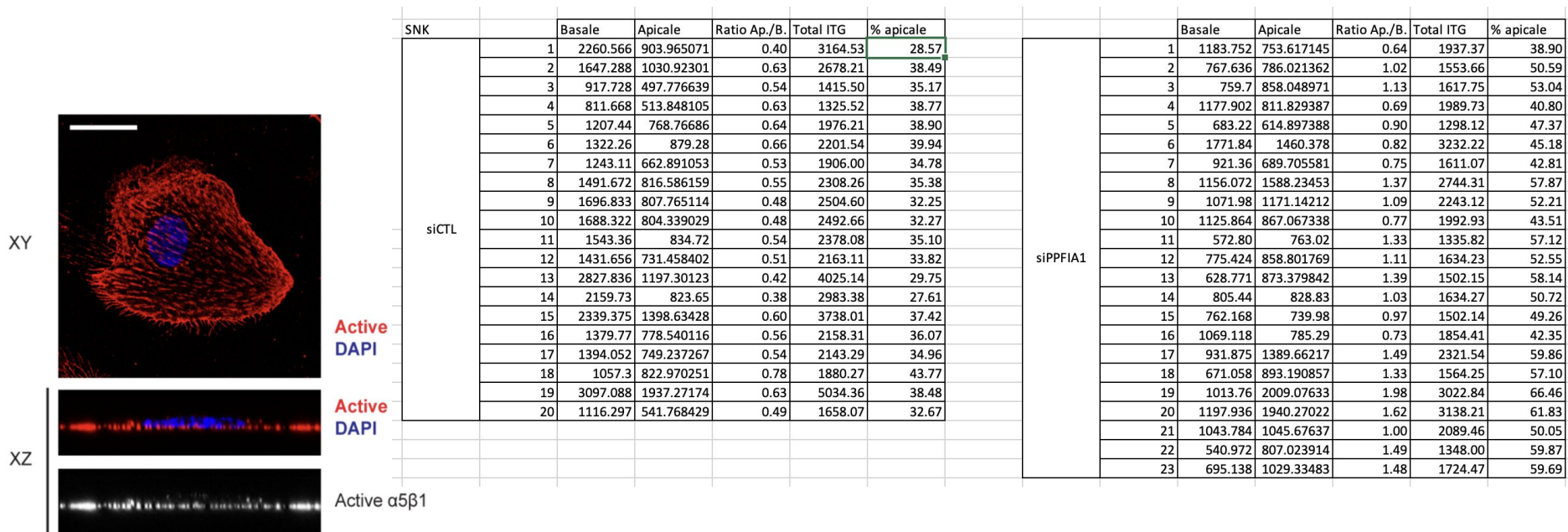
Example: amount of Integrins on the membranes

Examples of confocal microscopy experiment on xz axis



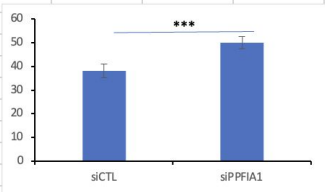
Example: amount of integrins on the **apical** membrane

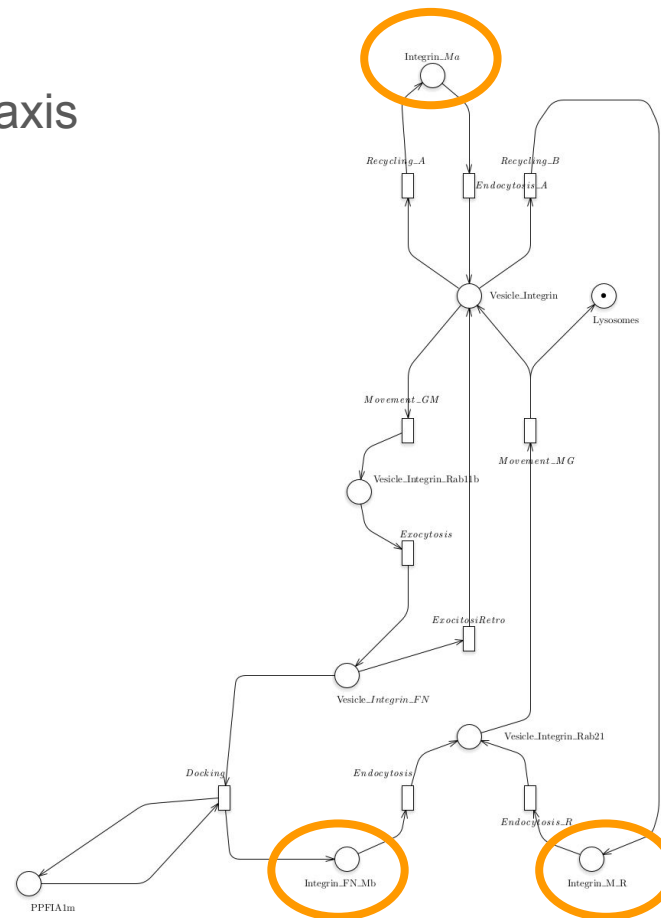
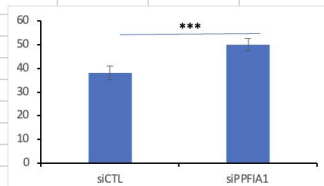
Examples of confocal microscopy experiment on xz axis



Example: amount of integrins on the apical membrane

Examples of confocal microscopy experiment on xz axis

Statistica complessiva							
Apicale		20201202	20201119	20201102	20201021	Media	SD
	siCTL	35.21	41.97	36.67	38.79	38.16	2.9375883
	siPPFIA1	52.06	50.81	46.09	51.31	50.07	2.7022162
		1.478541229	1.2104311	1.2568897	1.3229255		
							
Apicale	siCTL	siPPFIA1					
Media	38.158823	50.06520684					
SD	2.9375883	2.702216152					
TESTT	0.0004969	***					
Dato che in modello rappresenterò siCTL = 1, allora							
38,158 :1 = 50,065 :x							
	siCTL	siPPFIA1					
Itga5Itgb1Fn1nMa	1	1.31202178					
Da Madugundu 2019 normalizzato (vedi file Mass Spec), ottengo numero assoluto di quantità di proteina:							
495.071544							
Quindi:	siCTL	siPPFIA1					
	495.071544	649.54465					

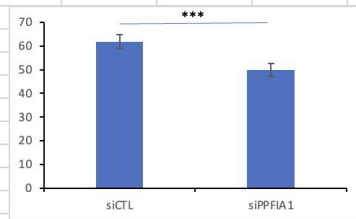


Example: amount of integrins on the basal membrane

Examples of confocal microscopy experiment on xy axis

Statistica complessiva							
Basale		20201202	20201119	20201102	20201021	Media	SD
	siCTL	64.79	58.03	63.33	61.21	61.84	2.93758833
	siPPF1A1	47.94	49.19	53.91	48.69	49.93	2.70221615
		0.739968653	0.84778308	0.85127631	0.79537364		
64.79:1=47.94:x							

Basale	siCTL	siPPF1A1
Media	61.841177	49.93479316
SD	2.93758833	2.702216152
TESTT	0.00049686	***

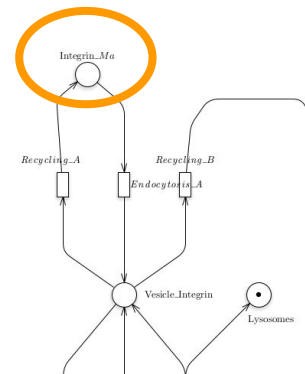


Dato che in modello rappresenterò siCTL = 1, allora
61,84 :1 = 49,934 :x

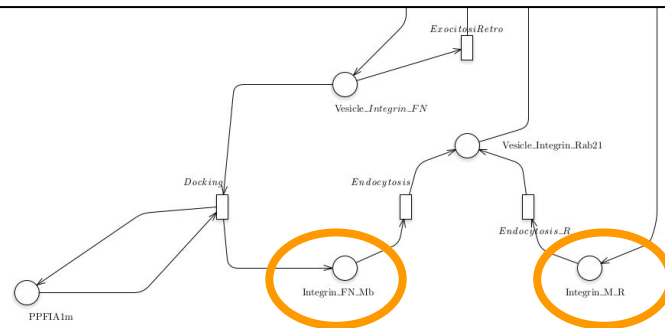
	siCTL	siPPF1A1
Itga5ltgb1Fn1nMa	1	0.80746835

Da Madugundu 2019 normalizzato (vedi file Mass Spec), ottengo
numero assoluto di quantità di proteina:
495.071544

Quindi:	siCTL	siPPF1A1
	495.071544	399.754603



Constraint:
Integrin_FN_Mb + Integrin_M_R = 495
tokens (t = 0h) OR 399.75 (t = 72h)

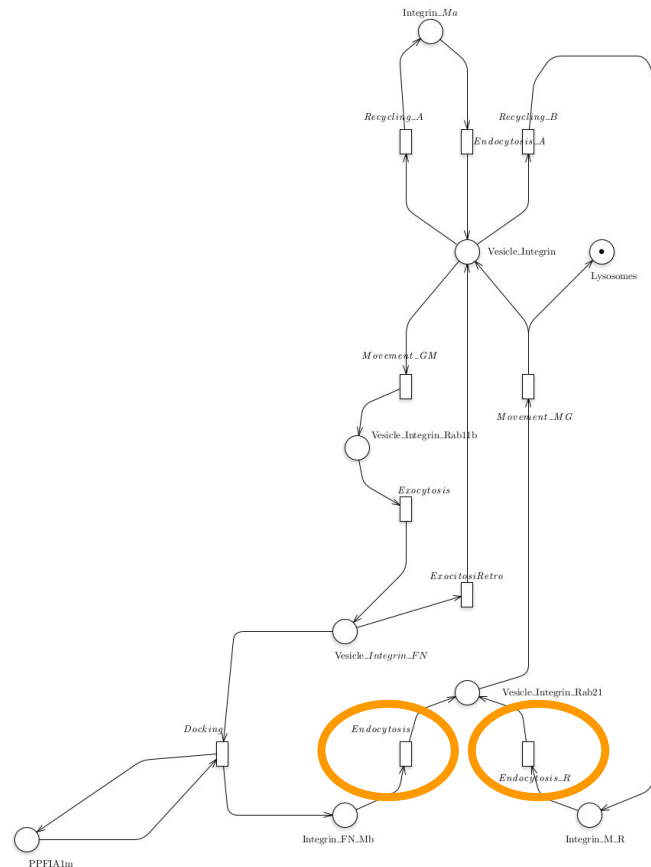
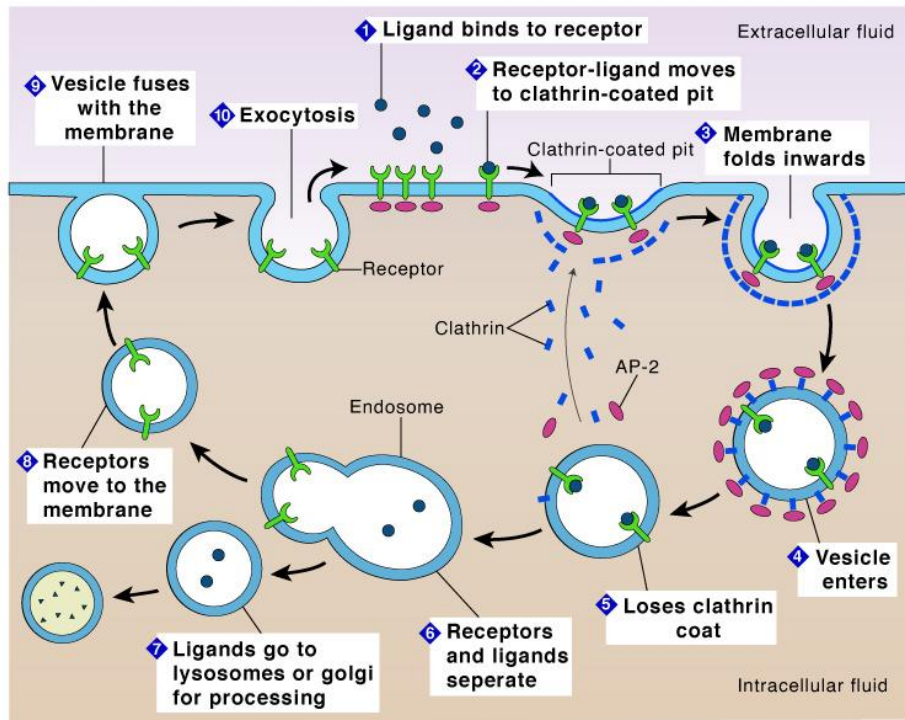


Example: parametrising a transition using wet lab data

Endocytosis:

Receptor-Mediated Endocytosis

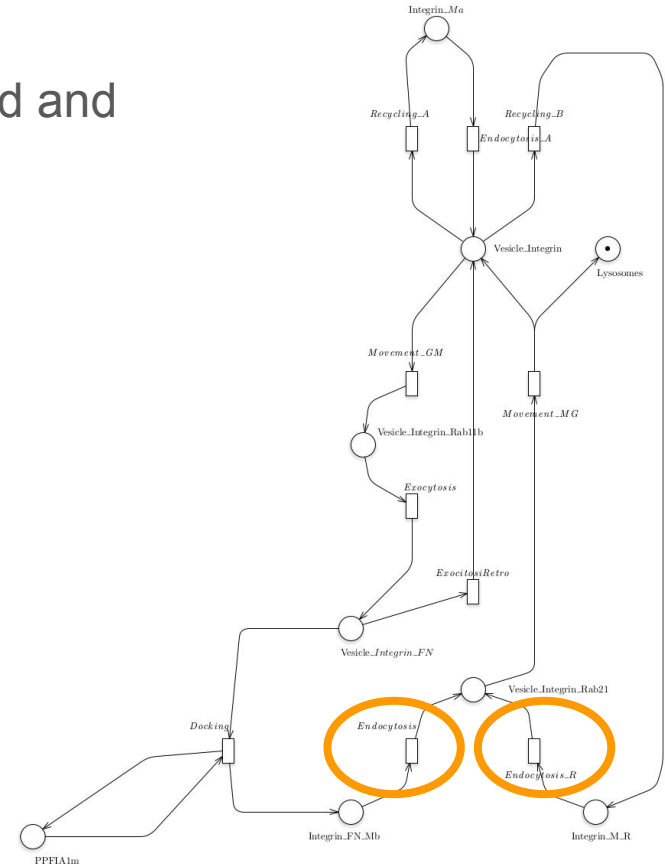
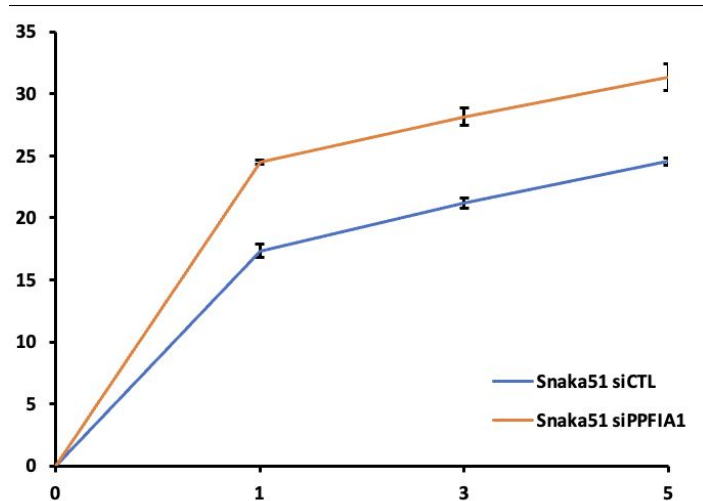
ScienceFacts.org



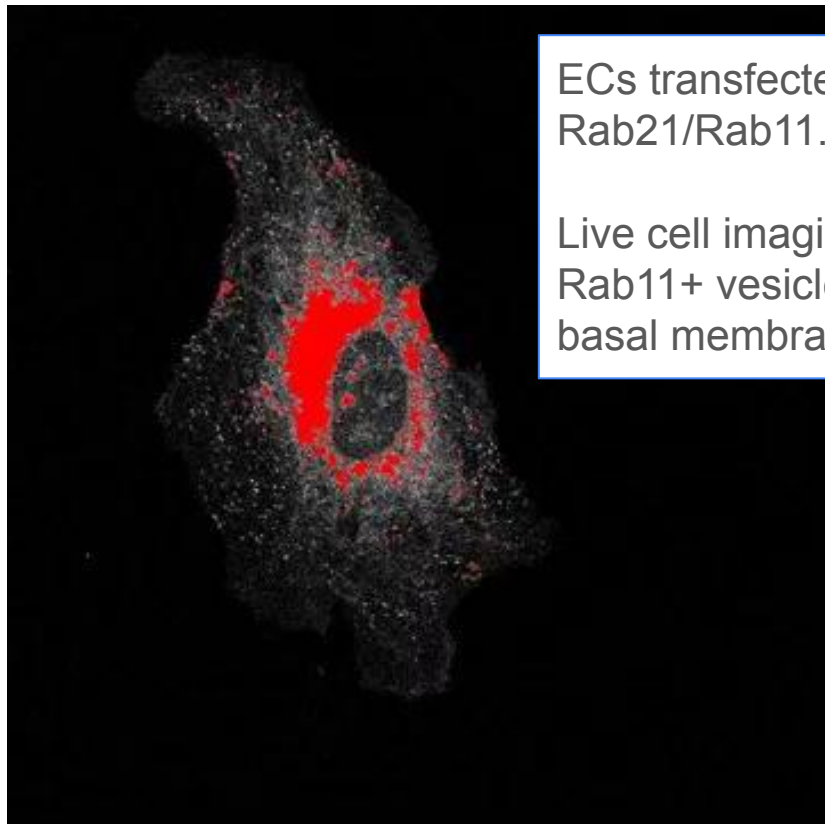
Example: parametrising a transition using wet lab data

Endocytosis experiment:
after 5 min, 25% of integrins have been endocytosed and
have endocytosis process has reached a plateau

Transition: $1/(5 \cdot 60)$ seconds

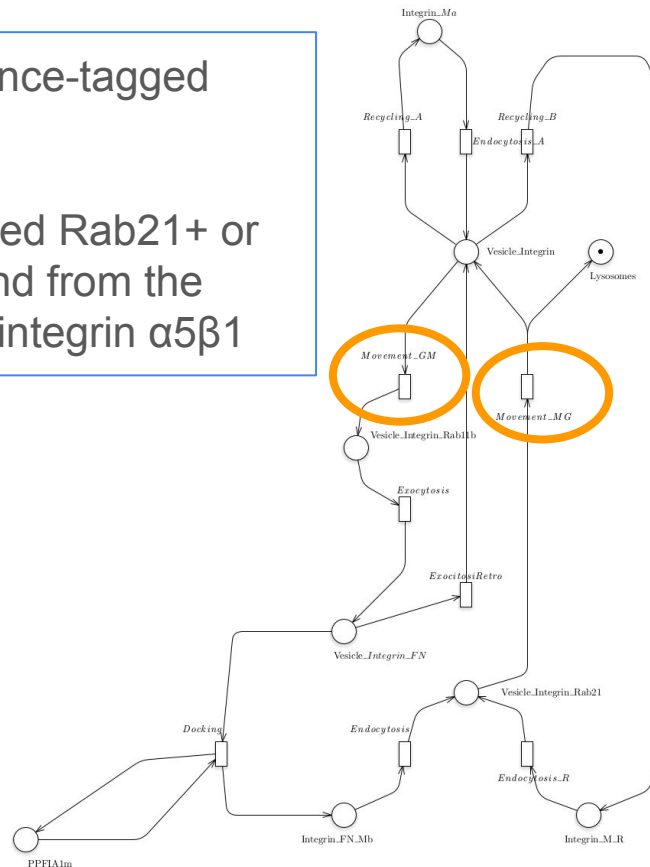


Example: parametrising a transition using wet lab data



ECs transfected with fluorescence-tagged Rab21/Rab11.

Live cell imaging videos detected Rab21+ or Rab11+ vesicles traveling to and from the basal membrane and carrying integrin $\alpha 5 \beta 1$

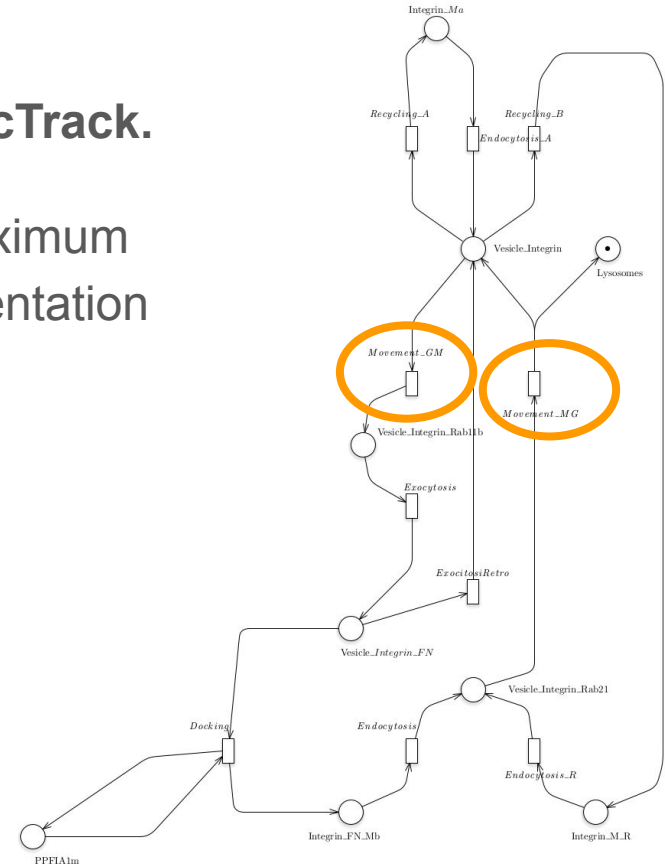


Example: parametrising a transition using wet lab data

We developed a MATLAB software code called **VescTrack**.

It exploits MATLAB built-in functions to segment maximum intensity projection images resulting in image segmentation and it allows users to automatically

1. identify and quantify intracellular vesicles,
2. track their movement in time, and
3. calculate their velocity.



Intracellular vesicles identification

Input: live cell imaging videos in which cells have been transfected with fluorescent Rab11 or Rab21, which mark vesicles

Vesicle segmentation:

1. Identification of **true signal** (= vesicles): determination of a threshold values corresponding to the average of pixel's intensities plus n times the standard deviation (value of n adjusted depending on the fluorescent intensity of the image)
2. Thresholding converts each frame of the movie into a **binary image** by automatically setting it to 1 (white) all those pixels whose values are higher than the threshold and to 0 (black) all the other pixels whose value is lower.
3. Spatially near-white pixels are then connected in order to form a **single object**
4. Morphology-based functions separate merged objects (vesicles must be round)
5. Keep objects with size in the range of 360 - 540nm (or 2-3 pixels)
6. **Label** objects

Intracellular vesicles tracking

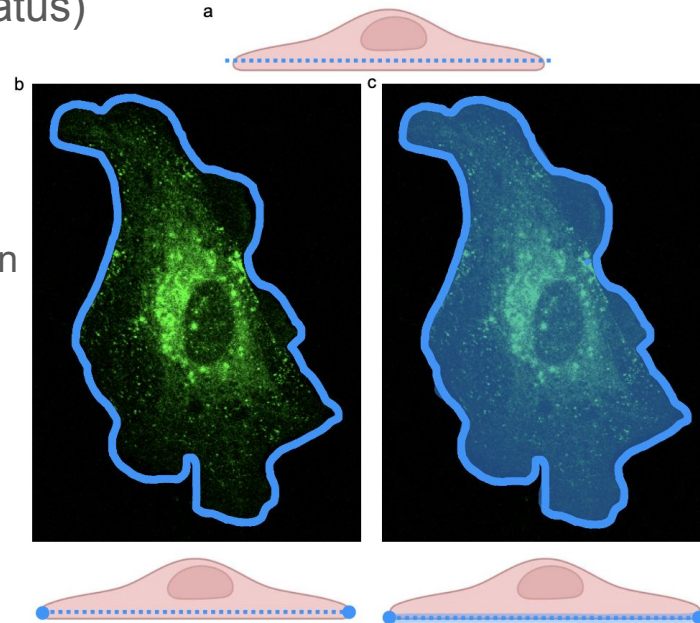
Tracking vesicles:

1. Store the spatial position of each object centroid

Intracellular vesicles tracking

Tracking vesicles:

1. Store the spatial position of each object centroid
2. Identify **starting** and **ending** points (perimeter of the cell or nucleus as proxy for Golgi apparatus)



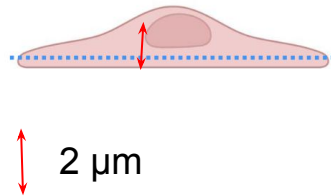
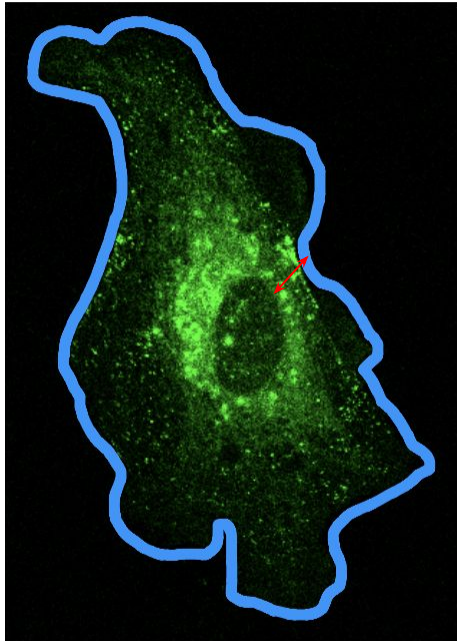
Perimeter: point of contact between basal and lateral membrane

Actual focal plane of the video (where most of the trafficking occurs) is above the basal membrane

Intracellular vesicles tracking

Tracking vesicles:

1. Store the spatial position of each object centroid
2. Identify **starting** and **ending** points (perimeter of the cell or nucleus as proxy for Golgi apparatus)



Vesicles that travel from/to perimeter to/from the nucleus are traveling the same distance as those that descend to basal membrane

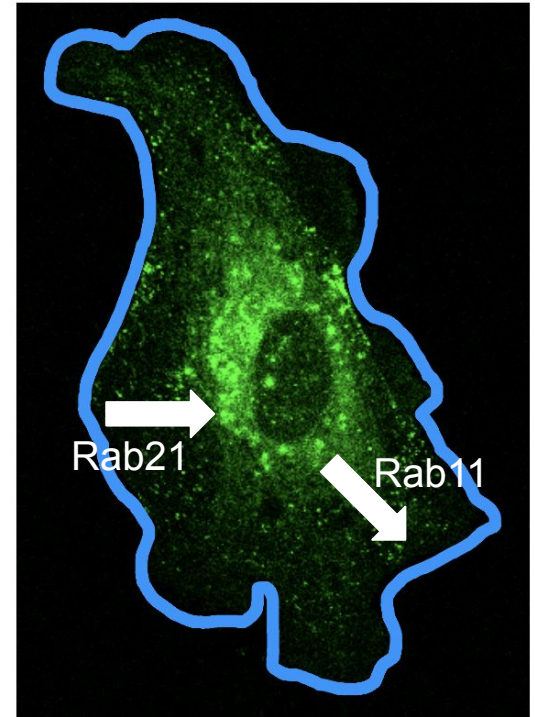
Intracellular vesicles tracking

Tracking vesicles:

1. Store the spatial position of each object centroid
2. Identify **starting** and **ending** points (perimeter of the cell or nucleus as proxy for Golgi apparatus)

Perimeter: when a black pixel was followed by at least 20 consecutive white pixels ± 3 pixels

Nucleus: a white pixel was followed by at least 4 consecutive black pixels ± 5 pixels



Rab21: basal membrane to Golgi
Rab11: Golgi to basal membrane

Intracellular vesicles tracking

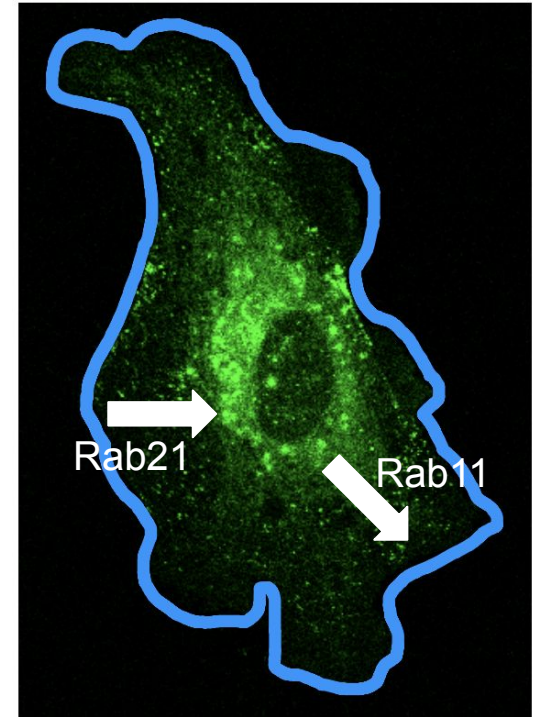
Tracking vesicles:

- Store the spatial position of each object centroid
- Identify **starting** and **ending** points (perimeter of the cell or nucleus as proxy for Golgi apparatus)

Perimeter: when a black pixel was followed by at least 20 consecutive white pixels +/- 3 pixels

Nucleus: a white pixel was followed by at least 4 consecutive black pixels +/- 5 pixels

- track all vesicles that start arrive/leave the **perimeter**
- track all vesicles that arrive/leave the **nucleus**



Rab21: basal membrane to Golgi

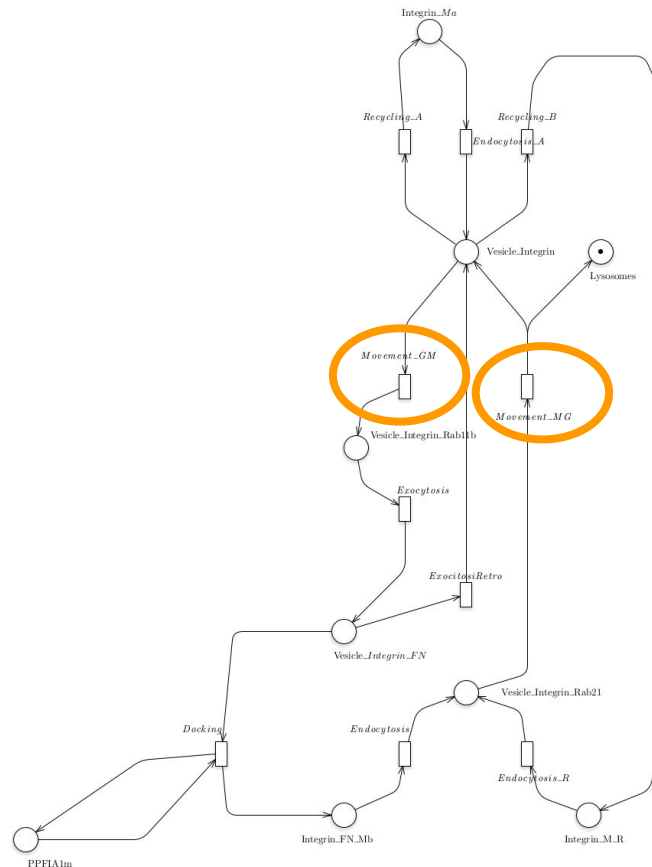
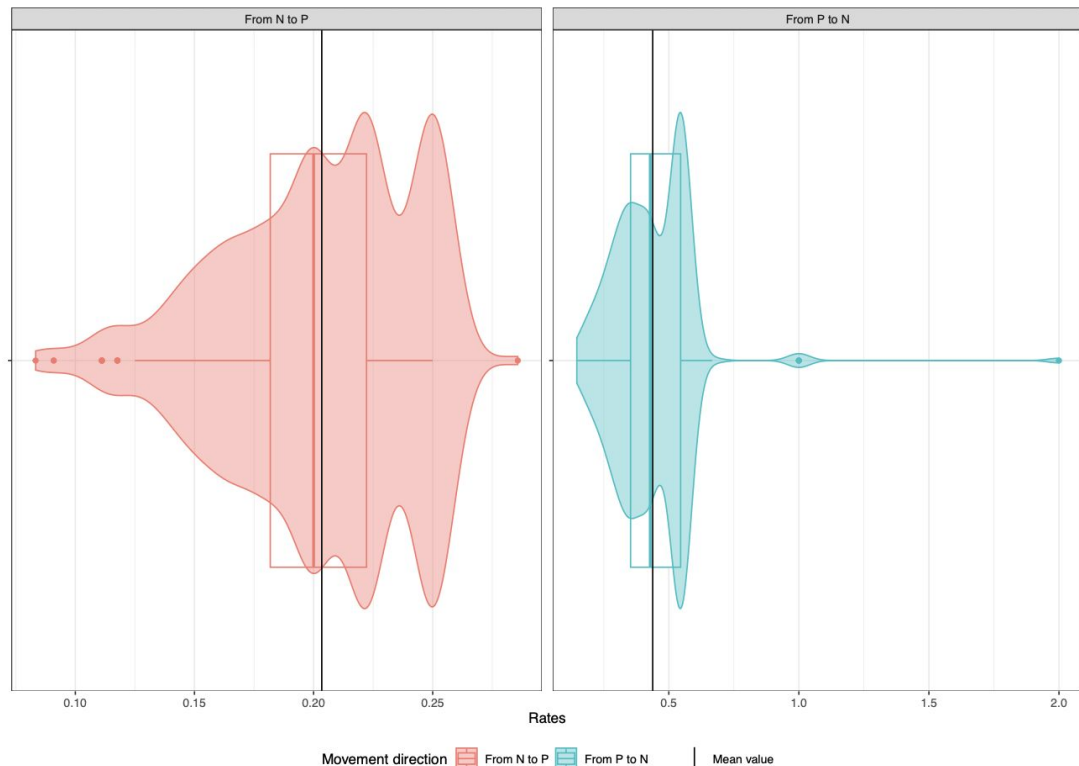
Rab11: Golgi to basal membrane

Tracking vesicles:

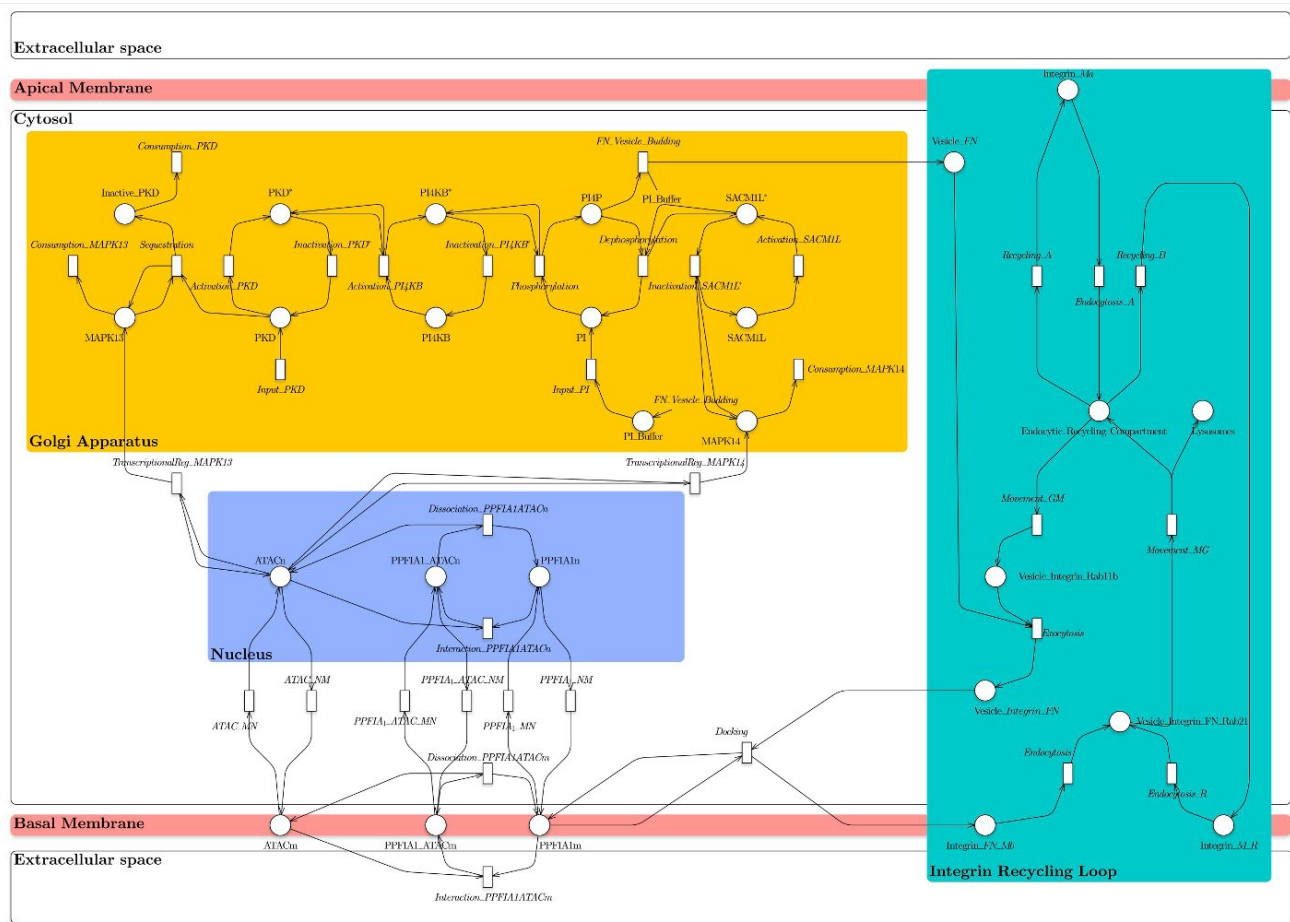
1. Store the spatial position of each object centroid
2. Identify **starting** and **ending** points (perimeter of the cell or nucleus as proxy for Golgi apparatus)
3. Tracking of vesicles over time: check frame by frame whether the position of each single object matches with any object of the previous frame within a range of ten pixels.
 - a. If more than one match is encountered, the closest to the one of the previous frame is accepted.
 - b. If no matches are found, the vesicle is considered as “disappeared”.
 - c. when a vesicle disappears and another one appears in its same position after two frames, the second one is considered a “new vesicle”.

Intracellular vesicles tracking

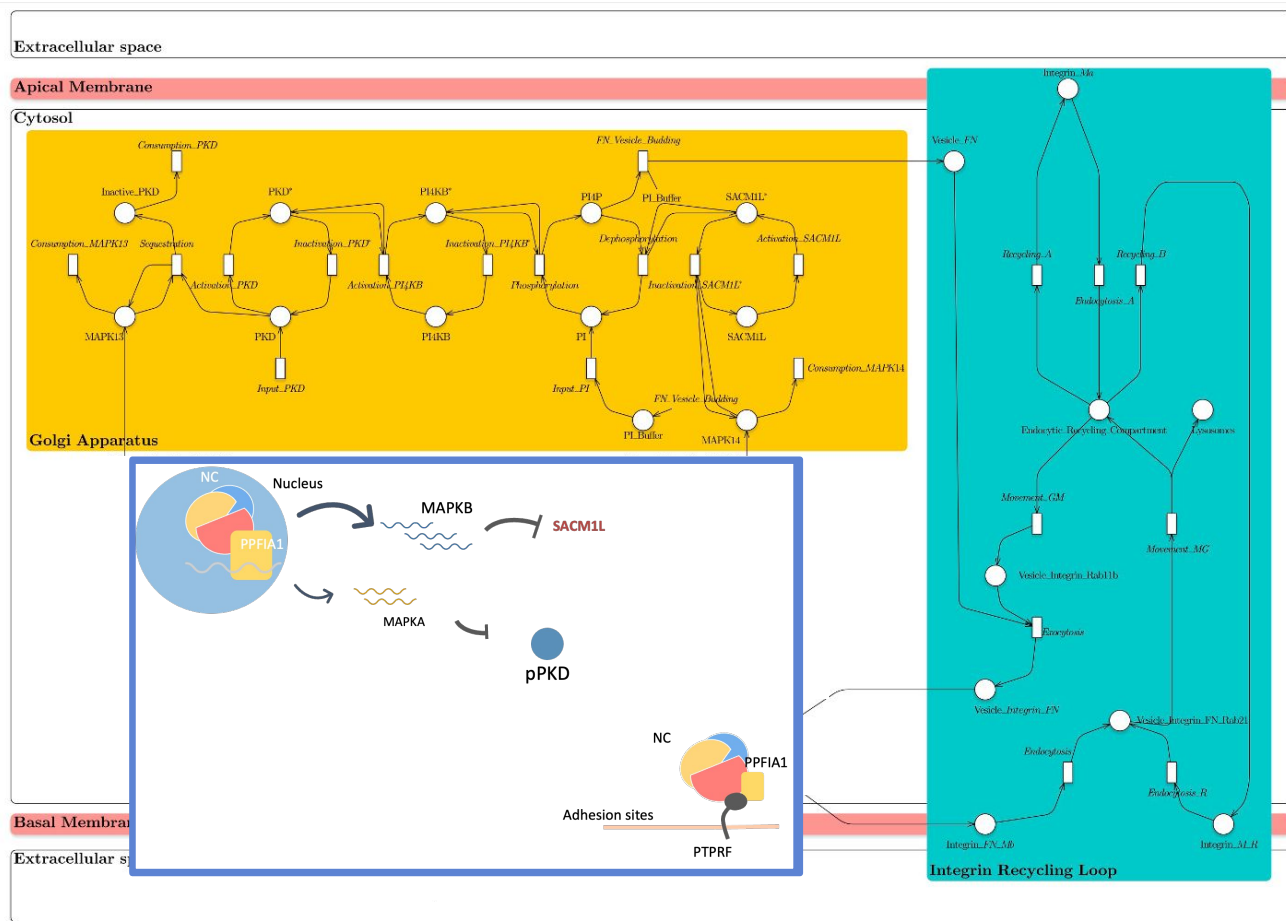
Result: transition velocities



Hands-on: drawing the PN of a missing portion of the model

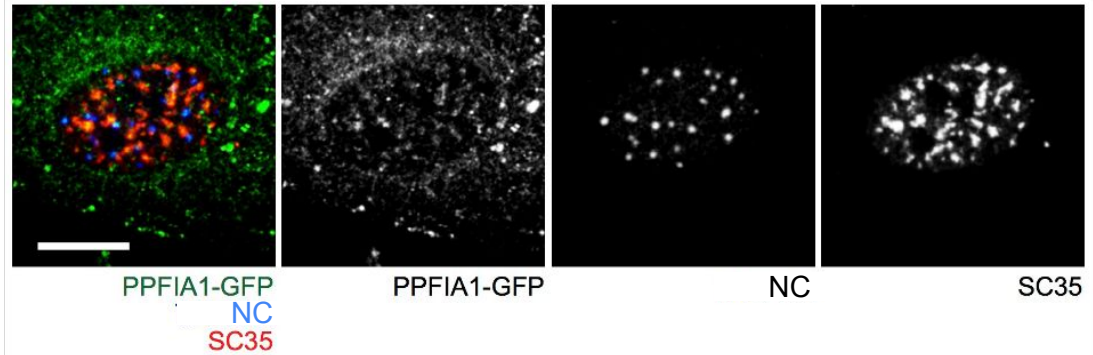


Hands-on: drawing the PN of a missing portion of the model

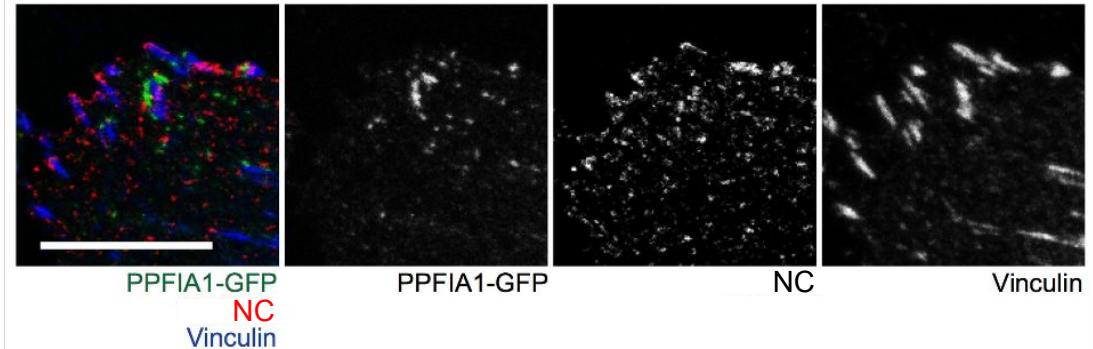


Given this data, how would you design your petri net?

1. PPFIA1 interacts with NC in the nucleus



2. PPFIA1 interacts with NC at adhesion sites



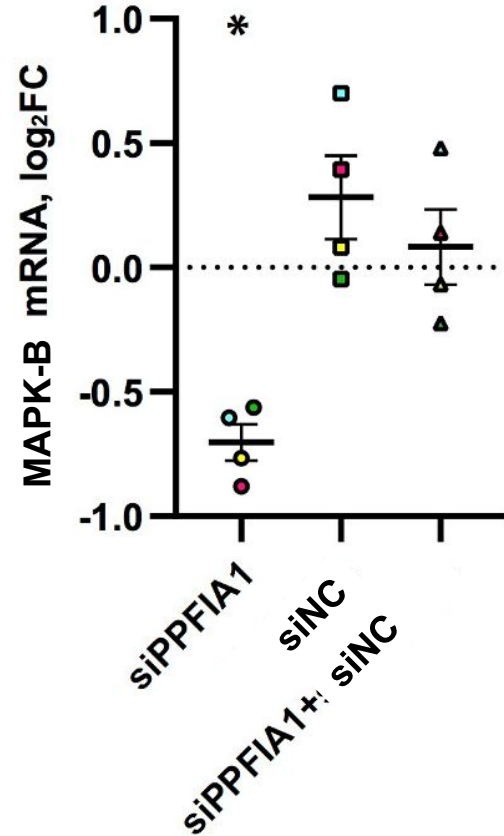
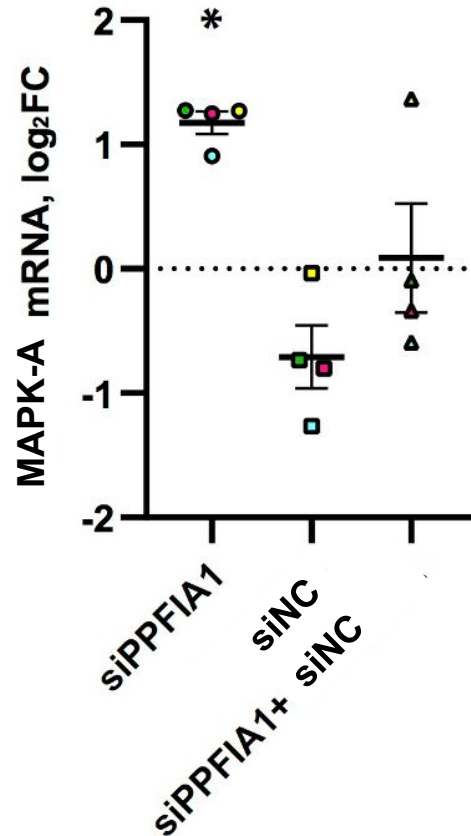
Given this data, how would you design your petri net?

3. PPFIA1 regulates MAPKA and MAPKB transcription *via* NC

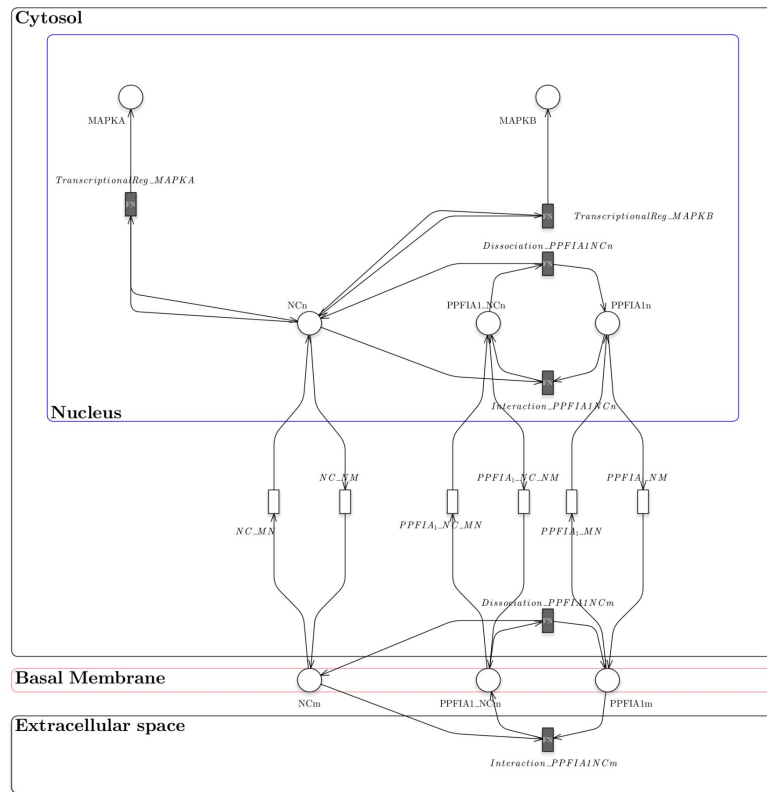


<https://github.com/qBioTurin/From-Data-to-Models.git>

Experimental replicates
#1 #2 #3 #4



Hands-on: drawing the PN of a missing portion of the model

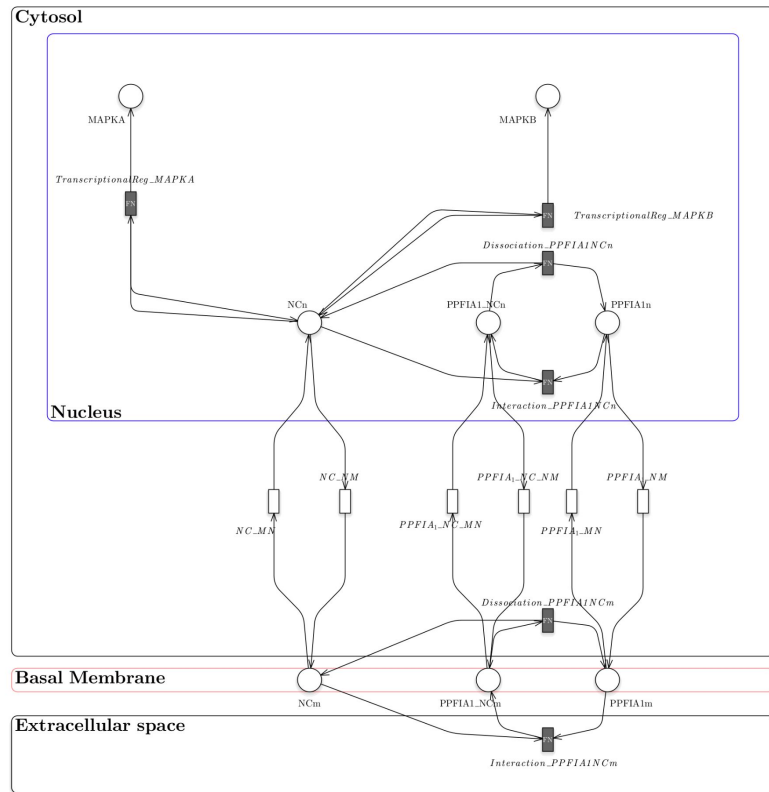


Let's parametrize our model

Model initial marking:

1. Add tokens to PPFIA1
2. Add tokens to NC
3. Add tokens to MAPKA and MAPKB

And final markings!



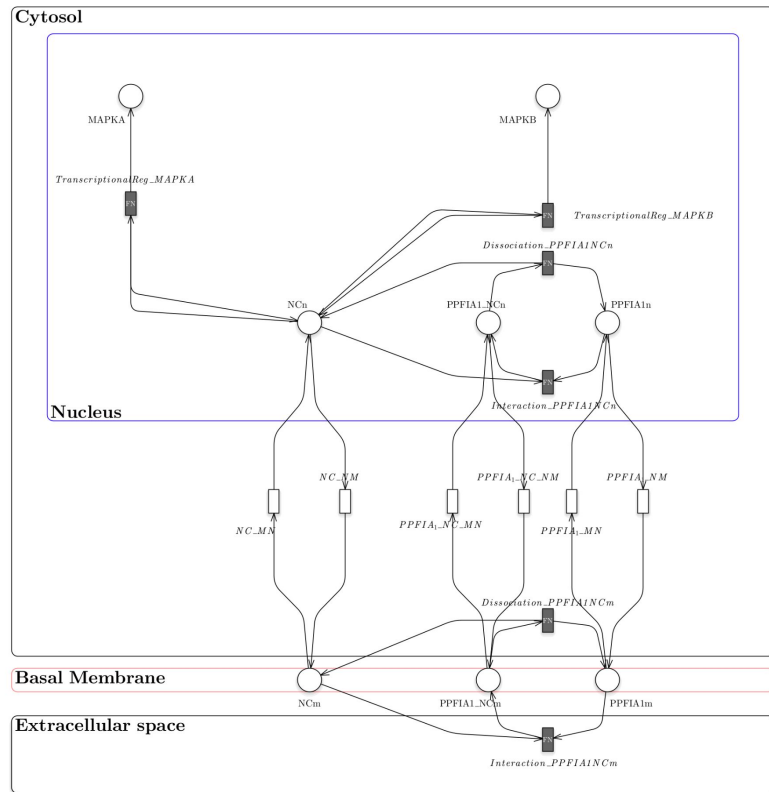
Let's parametrize our model

Model marking:

1. Add tokens to PPFIA1
2. Add tokens to NC
3. Add tokens to MAPKA and MAPKB

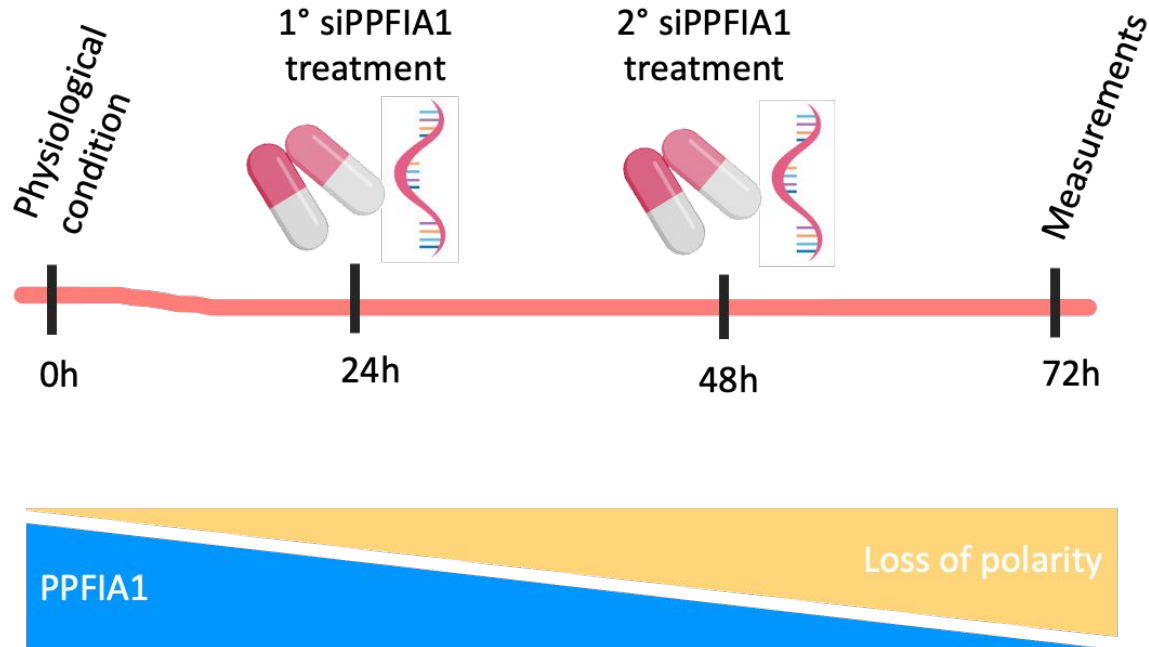
Transition rates:

1. Removing PPFIA1 at set times
2. Interaction and movement rates: to be calibrated
3. Transcription rates



Progressively removing PPFIA1

We are modeling a single endothelial cell as
PPFIA1 is progressively removed by siRNA.



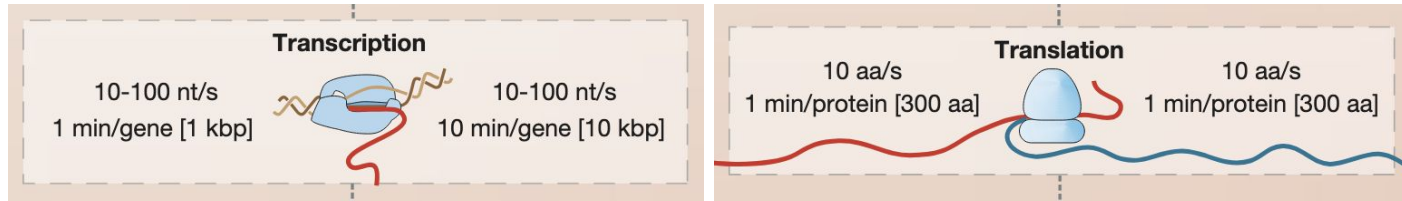
Creating a function to simulate PPFIA1 removal

```
EventFun = function(marking,time){  
  newmarking <- marking  
  names(newmarking) <- readRDS("/home/docker/data/Input/NAMES.RDS")  
  
  if (time == 24*60*60){  
    PPFIA = 3.21 * 0.138  
  }else if(time == 48*60*60){  
    PPFIA = 0.1  
  }  
}
```

Parametrizing transcription reaction

MAPK-A and MAPK-B transcriptional regulation reactions have to take into account two aspects:

1. temporal frame in which the biological processes of transcription and translation occur → “rate_expression”



[https://www.cell.com/cell/pdf/S0092-8674\(16\)30208-2.pdf?_returnURL=https%3A%2F%2Flinkinghub.elsevier.com%2Fretrieve%2Fpii%2FS0092867416302082%3Fshowall%3Dtrue](https://www.cell.com/cell/pdf/S0092-8674(16)30208-2.pdf?_returnURL=https%3A%2F%2Flinkinghub.elsevier.com%2Fretrieve%2Fpii%2FS0092867416302082%3Fshowall%3Dtrue)

MAPK-A: 6316 nt and 365 AA ~450 seconds → rate_expression_A = 1/450s

MAPK-B: 4222 nt and 360AA ~325 seconds → rate_expression_B = 1/325s

2. modulation of gene transcription by NC activity → “cost_modulation” → estimated

Parametrizing transcription reaction

MAPK-A and MAPK-B transcriptional regulation reactions have to take into account two aspects:

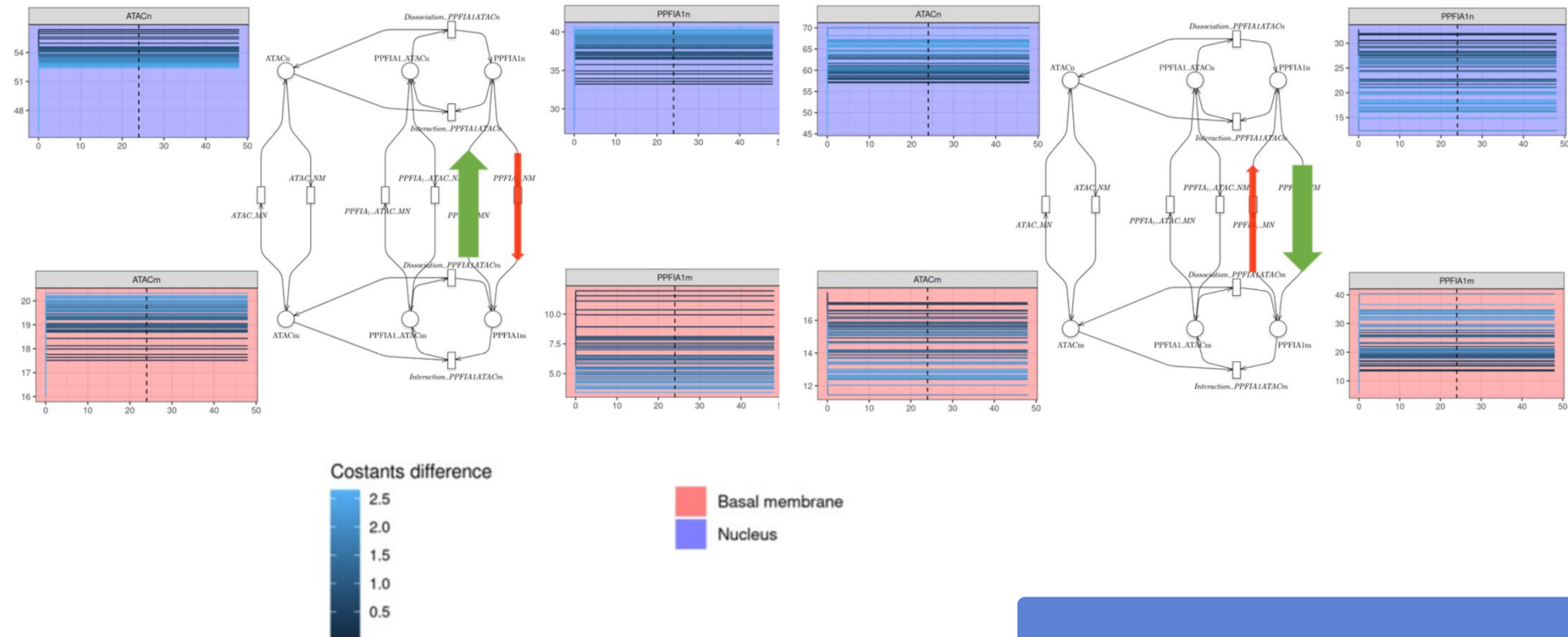
1. temporal frame in which the biological processes of transcription and translation occur → “rate_expression”
2. modulation of gene transcription by NC activity → “cost_modulation” → estimated

$$f(\text{TranscriptionalReg_MAPK_A}) = \text{rate_expression_A} + x_NC * \text{cost_modulation_A};$$

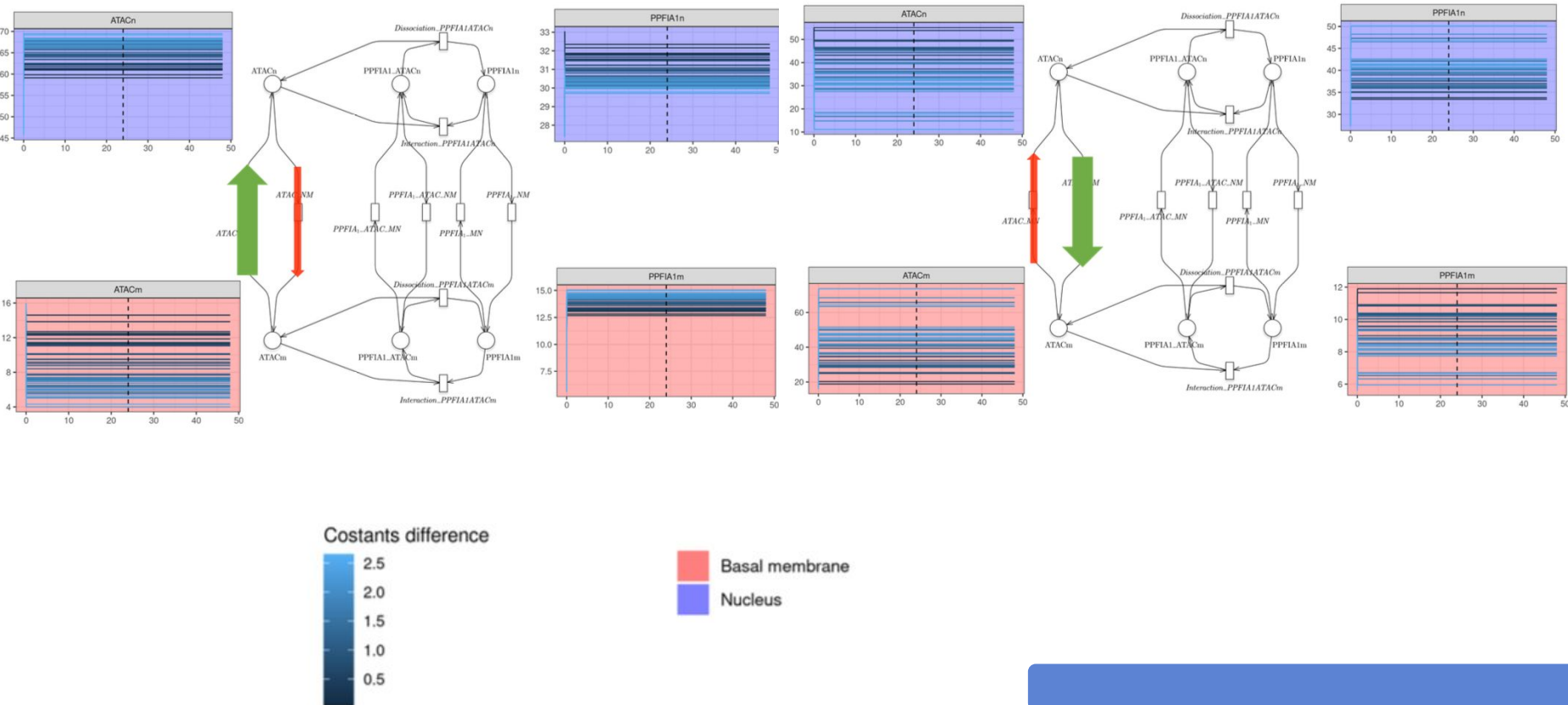
$$f(\text{TranscriptionalReg_MAPKB}) = \text{rate_expression_B} - x_NC * \text{cost_modulation_B},$$

$$\text{cost_modulation} \geq \text{rate_expression_B} / \max(x_NC).$$

Hands-on: drawing the PN of a missing portion of the model

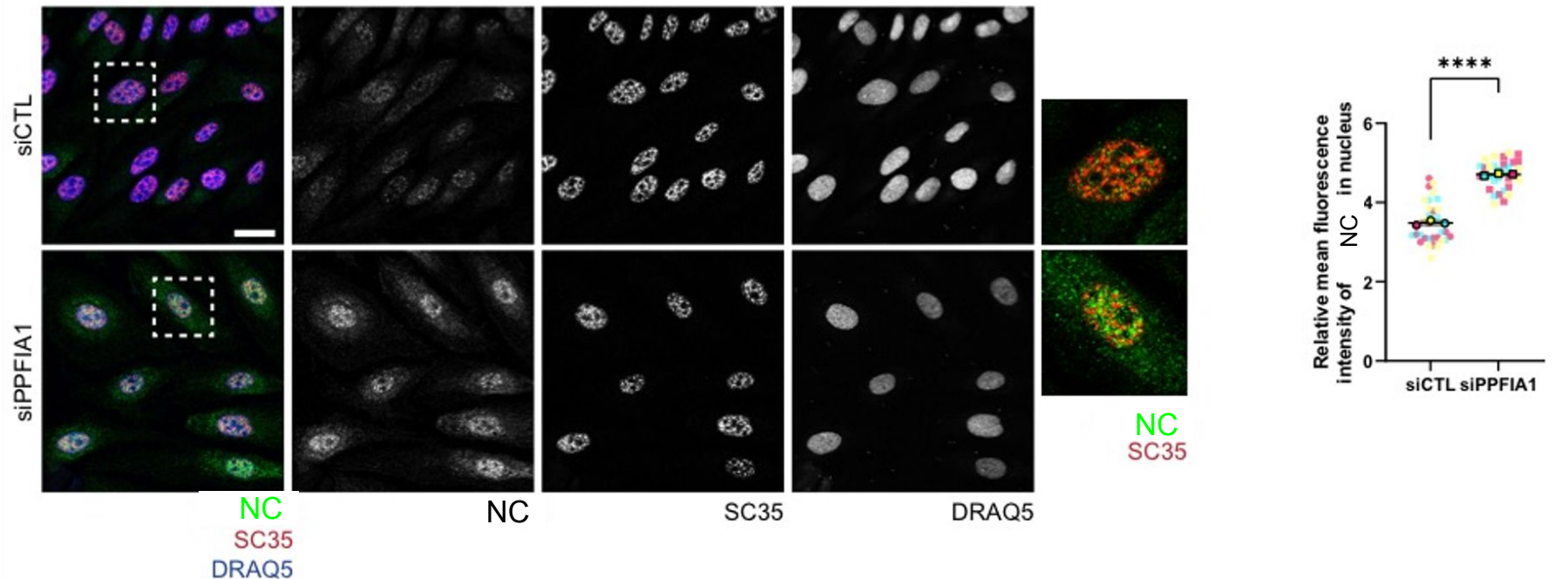


Hands-on: drawing the PN of a missing portion of the model



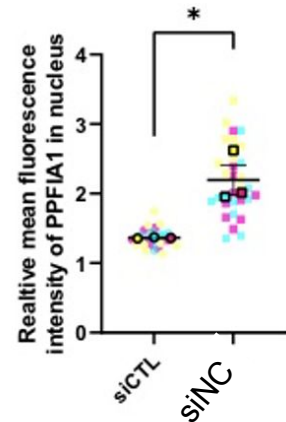
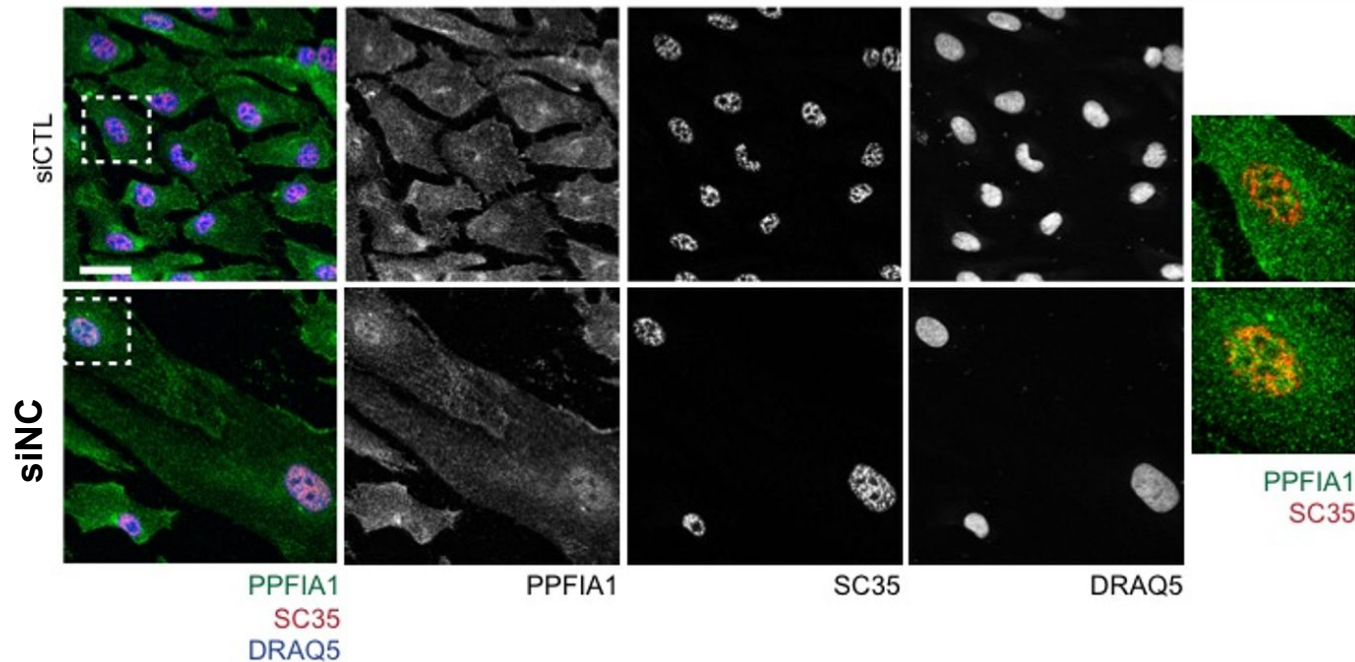
Hands-on: drawing the PN of a missing portion of the model

Hypothesis confirmation in the lab:
PPFIA1 physically and functionally inhibits NC complex

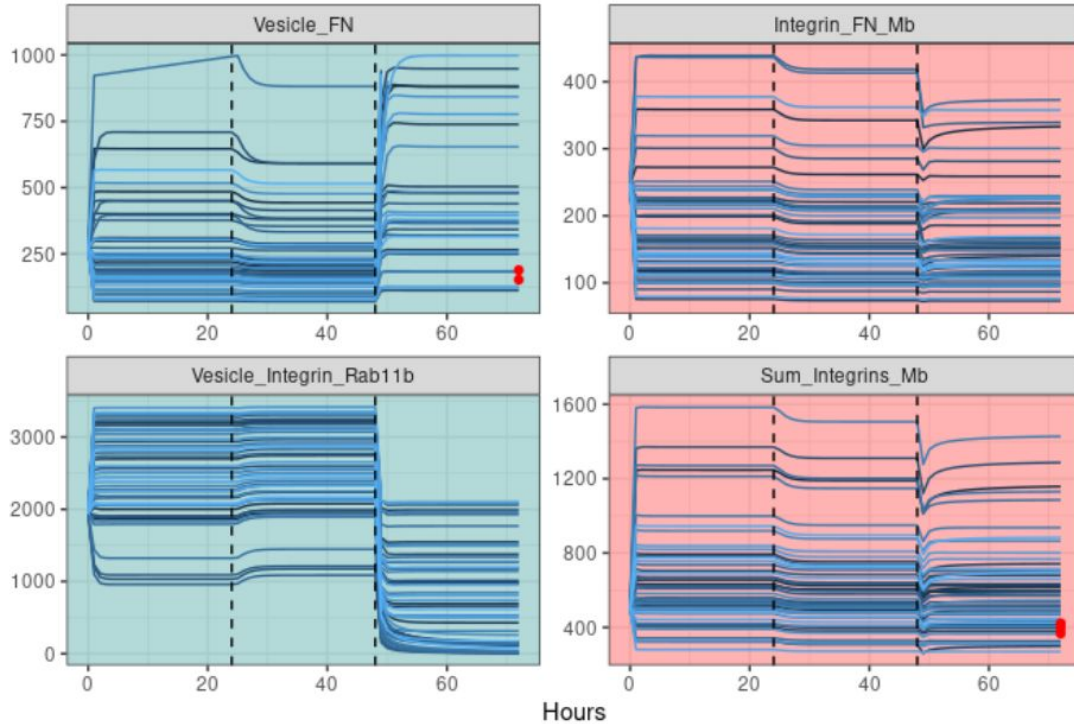


Hands-on: drawing the PN of a missing portion of the model

Hypothesis confirmation in the lab:
PPFIA1 physically and functionally inhibits NC complex



Third step: model calibration



Relative Mean Error

$$\text{RME} = \frac{1}{n} \sum_{i=1}^n \frac{|y_i - \tilde{y}_i|}{\tilde{y}_i}$$

y_i = simulated datum

$\tilde{y}_i$ = observed datum

Fourth step: model simulation and analysis



<https://github.com/qBioTurin/From-Data-to-Models.git>